

A Multimodal Model for Predicting Feedback Position and Type During Conversation

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Abstract

This work investigates conversational feedback, that is the specific reaction of a listener in response to a speaker, occurring during all natural interaction. Feedback does not occur randomly. It depends on the main speaker's discourse and in return supports its elaboration. As a consequence, the quality of interaction highly depends on feedback production.

We aim to consider feedback in its entire complexity, firstly by studying it according to two main types of feedback, *generic* and *specific* (while most works in the literature focus only on generic types). Secondly, we propose a new fine-grained labeling system by introducing two sub-types of specific feedback: *positive/negative* and *expected/unexpected*. We then experiment a set of multimodal features in order to automatically predict on the one hand, the possible position of the feedback, and on the other hand, its type. This computational modeling makes it possible to precisely identify the different features from the speaker production (morpho-syntactic, prosodic and mimogestural), that play a role in triggering the listener feedback, and also to evaluate their relative importance in predicting that feedback. By integrating all the different modalities as well as high-level features, our predictive model leads to good scores in comparison with more focused works.

The main contribution of this work is threefold: 1/ A substantial literature review in both linguistic and machine learning field highlighting the complexity factors for feedback prediction 2/ the characterization of a new set of relevant multimodal features for predicting feedback and 3/ a computational model for the automatic prediction of all types of feedback. The present model, applied to french corpora, constitutes the state of the art in this task.

Keywords : Feedback, Multimodality, Linguistic interaction, Statistical model, Corpus study.

1. Introduction

Conversational interaction is a dynamic and systemic activity during which participants collaborate in order to establish an exchange, based on shared information and mutual knowledge. The *dynamic* aspect results from the fact that participants frequently change roles, switching from the speaker (who holds the floor) to the listener (who is actively listening). The *systemic* aspect refers to the implicit rules that control the interaction and ensure its success [Sacks et al., 1974]. These rules concern many different aspects (turn-taking, lexical choice, pace, prosody, body posture, etc.) which also reveal participant engagement [Glas and Pelachaud, 2015]. Conversational feedback is among the most important of such phenomena [Schegloff, 1982]. It is classically described as a brief signal produced by the listener in response to the speaker's speech. Feedback (also called *backchannels*) can be verbal (e.g. *yes*), vocal (e.g. *mhm*), and/or gestural (head movements, eyebrows movements, smiles). By using feedback, the listener gives the speaker information about her/his awareness, comprehension and appreciation, helping him/her in the elaboration of his/her discourse. Dialog studies underline the crucial role of feedback for updating shared knowledge (also called *common ground*) [Clark, 1996, Horton, 2017] and promoting the alignment between participants which is a pre-requisite for a mutual comprehension and more generally for achieving successful interaction [Bavelas et al., 2000, Pickering and Garrod, 2013, Pickering and Garrod, 2021].

In the perspective of human-machine interaction, feedback production (in terms of quantity, type and timing) is central to the perception of an artificial agent as well as for the engagement of the participants [Poppe et al., 2010, Poppe et al., 2011, Truong et al., 2011, Glas and Pelachaud, 2015]. An artificial

agent is needed to produce appropriate responses (in terms of feedback and/or turn-taking) to the speaker's lexical content by quickly analyzing changes and displaying rapid adaptation. Thus, it is of crucial importance to better understand and model human feedback behavior in human-human / natural context to be able to automatically predict the artificial agent feedback for improving human-machine interaction.

Several works based on corpus analysis have shown that the production of feedback can be triggered by different multimodal cues from the speaker's speech [Koiso et al., 1998, Ward and Tsukahara, 2000, Allwood and Cerrato, 2003], sometimes also called *inviting-cues* [Gravano and Hirschberg, 2011].

Different computational models have been proposed to automatically predict feedback. Nowadays, they are mainly based on a machine learning approach from a conversational corpus. In most cases, the models focus either on *verbal* feedback [Okato et al., 1996, Ward and Tsukahara, 2000, Cathcart et al., 2003] or *gestural* feedback [Morency et al., 2010, Ozkan and Morency, 2010]. Very few works have considered both gestural and verbal feedback prediction [Fujie et al., 2004, De Kok et al., 2010]. Moreover, existing models usually only focus on the most general type of feedback such as *mh*, *yeah* or a nod. Only a few works have recently started to study more complex feedback [Kawahara et al., 2016, Ortega et al., 2020, Jang et al., 2021].

A first source of difficulty in conducting this work is that it requires conversational corpora with a wide set of multimodal annotations not only used to characterize the feedback itself (in the listener's production) but also to obtain the different features for prediction (in speaker production). A second source of difficulty is the extreme variability of feedback production. Models predict potential sites of feedback. Nonetheless, not all of such sites are currently used to produce feedback [Heldner et al., 2013, Howes and Eshghi, 2021], making it difficult to evaluate the models. Standard machine learning metrics (f-score, precision and recall) are good indicators for finding the best set of features and for testing their contribution. However, perfect matching between predicted feedback and observed feedback is almost impossible due to this high variability in feedback production, resulting in relatively low scores.

In this paper, we propose to consider all types of feedback, leading for the first time to a complete model of feedback production. Following [Bavelas et al., 2000], we distinguish between *generic* and *specific* feedback. The former are often items such as *mh* or a nod and express interest and understanding of the current discourse while the latter are more complex (verbal utterances, tone of voice, etc) and involve a more evaluative function. We add two subtypes (negative/positive and expected/unexpected) to the specific category allowing us to cover all uses of the feedback in our corpus [Boudin et al., 2021]. We present a hierarchical method to automatically predict firstly the feedback positions and then their types, based on multimodal (verbal and visual) features. The first task concerns the temporal prediction, or more precisely the prediction of the potential sites of feedback. The second task aims at classifying the feedback types according to general types (generic and specific) and sub-types (negative/positive; expected/unexpected). To the best of our knowledge, this is the first approach attempting to predict both feedback positions and precise types.

Our work is based on three main hypotheses. Firstly, that taking into account a rich set of multimodal features involving high-level linguistic features from different modalities will improve the performance of the model. Secondly, we also think that this type of modeling, based on a rich set of features, will also improve the interpretability of the model: our features being linguistically motivated, we know the type of information they encode, which makes the interpretation transparent. Thirdly, we claim that for the classification of feedback types, a different set of relevant features will emerge. Technically, our approach is based on both conversational analysis and machine learning methods considering multimodal cues.

The paper is organized as follows: in Section 2, we present the definitions of the types of feedback and the inviting-cues which have been identified in the literature. Section 3 overviews the main feedback predictive models proposed in the literature, distinguishing two types of task: continuous prediction in 3.1 and non-continuous prediction in 3.2. Section 4 contains the proposed feedback typology (section 4.1), a description of the data (section 4.2), a sub-corpus of the French multimodal corpus PACO and Cheese! [Amoyal et al., 2020, Priego-Valverde et al., 2020], their distribution in the data (section 4.3), and the extracted features (section 4.4). Models are then presented in Section 5 and 6, with the description of the method used, the obtained results and discussions for each type of task (continuous

prediction and feedback type classification). Finally, section 7 discusses our main results and presents our perspectives.

2. Feedback and their Predictive Cues: Theoretical Background

Before entering into the modeling aspects, it is necessary to specify the notion and the nature of feedback and to identify the different types of verbal and non-verbal features from the main speaker production that have been proposed in the literature as predictive cues for feedback production.

2.1 Feedback in the Literature

The notion of *feedback* was introduced a long time ago [Schegloff, 1982] in the perspective of underlining the collaboration between a speaker and an interlocutor during an interaction. Feedback is multimodal and can have different forms: nods, smile, laugh, short verbalization, facial expressions, eyebrow movements, hand gestures etc. This influential contribution, [Schegloff, 1982] highlights the different functions of these types of feedback. Firstly, they play the role of *continuers*, showing an interlocutor's interest and comprehension of the communicative situation. They can also show disinterest for example by repeating the same feedback for many utterances without any change. Feedback can also express a reaction (surprise, disgust, happiness) related to the semantic content. In this case, their forms are more variable, generally longer and more often lexicalized than *continuers* (e.g. "oh my god", "oh wow", "really") and they perform different functions such as *assessment*, *acknowledgment*, etc.

In another important contribution, [Bavelas et al., 2000] introduced a typology of feedback response, pointing out the active implication of interlocutors during interactions (especially during narration). The authors showed how such linguistic and para-linguistic behavior on the part of the listener has an impact on the speaker production. According to [Bavelas et al., 2000], interlocutors can produce two types of feedback: **generic** and **specific** feedback, similar to the distinction between continuer and assessment. Generic responses preferentially occur in the first part of the narration, within the set-up phase and the construction of the common ground. Such responses mostly include nods and/or short vocalizations (e.g. "yeah", "mhmh", "okay"). However, specific responses are closely connected to the semantic context. They tend to occur later in the narration, once the common ground has been established. Therefore, the interlocutor has enough information to react with a particular element (wince, exclamation, rising tones) that can show surprise, amusement, enthusiasm, etc. Several studies [Stivers, 2008, Tolins and Fox Tree, 2014, Bertrand and Espesser, 2017] have confirmed the relevance of such a typology.

2.2 Feedback Predictive Cues

Many different works have been produced to identify the different cues that play a role in the prediction of feedback. We propose in this section to summarize the main features, by modality.

Prosodic cues: One of the first studies focusing on a feedback inviting-cue proposed by [Ward, 1996] concerns the prosodic level in particular. The author demonstrated the importance of a low-pitch period in the speaker's speech for identifying the appropriate place to generate feedback. This period lasts at least 110 milliseconds. In addition, [Ward and Tsukahara, 2000] proposed rules for predicting feedback by identifying five conditions, as described in Figure 1. Moreover, the authors indicated that low-pitch regions are associated with no new information, which tends to promote the production of generic feedback¹.

¹ It is worth noting that we are making the distinction between the generic and specific terms of feedback according to the description made in cited papers, in order to make the paper more intelligible.

Upon detection of:

- (P1) a region of pitch less than the 26th-percentile pitch level and
 - (P2) continuing for at least 110 milliseconds,
 - (P3) coming after at least 700 milliseconds of speech,
 - (P4) providing you have not output back-channel feedback within the preceding 800 milliseconds,
 - (P5) after 700 milliseconds wait,
- you should produce back-channel feedback.*
-

Figure 1: [Ward and Tsukahara, 2000] hand-craft rules for feedback generation.

[Koiso et al., 1998] investigated the role of intonation, energy and duration cues as follows:

- Duration of the final phonemes over all IPUs: *short, normal or long*.
- F0 pattern: *rising, falling or flat*.
- F0 peak: *high or low* (above or below the center of the F0 range of the speaker)
- Energy pattern: *decreasing, late-decreasing, non decreasing*
- Energy peak: *high or low* (above or below the center of the energy range of the speaker)

They found falling, flat-fall and rise-fall F0 pattern, late decrease of energy and high peaks of energy to be relevant predictive cues. On the contrary, they also found that flat F0 pattern, short duration of the final phonemes, no decrease of energy, and low peak of energy discouraged feedback production.

[Poppe et al., 2010], found that speaker pauses longer than 400ms indicated a relevant place to produce feedback. The latter obtained better results by associating pauses with regions of low-pitch. Furthermore, [Terrell and Mutlu, 2012] found that pause durations of 500ms promoted feedback production.

Feedback predictive cues found by [Gravano and Hirschberg, 2011] were final-rising intonation (high rise and low rise) over the last 200-300ms of the previous IPU (Inter Pausal Unit), IPUs with longer duration, lower noise-to-harmonics ratio in the last 500-1000ms of the previous IPU and a highest pitch and intensity mean.

More recent work by [Brusco et al., 2020] confirmed the previous results. Effectively, they found a set of prosodic cues for feedback prediction consisting of IPU duration, final-word lengthening, speech rate, high-rising final intonation, plateau intensity level, and noise-to-harmonics ratio.

Morpho-syntactic cues: [Ward and Tsukahara, 2000] investigated the role of utterance endings, completion of grammatical clause, clause connective, disfluency markers, and sentence final particles (e.g. "you see") that seemingly promoted feedback production. Moreover, they underlined that these elements often co-occur with low-pitch regions.

[Bertrand et al., 2007], indicated that feedback appears most of the time after a noun, a verb or an adverb, but rarely after a determinant. Similar results were obtained by [Gravano and Hirschberg, 2011], when a set of part-of-speech (POS) bigrams were tested. Out of the 10 bigrams tested, 3 seemed to be significantly more present before feedback: determinant-noun ; adjective-noun and noun-noun.

Mimo-gestural cues: In their work, [Allwood and Cerrato, 2003] investigated gestural feedback. They found in particular a certain type of alignment between interlocutors during an interaction. Interlocutors tended to reproduce head movements produced by the speaker as gestural feedback. Speaker nods also appeared to be important in [Terrell and Mutlu, 2012, Stivers, 2008] for bimodal feedback that are produced in overlap. Note that laughs, smiles and such attitudes are not often investigated in the literature as feedback-predictive-cues.

In [Poppe et al., 2010, Terrell and Mutlu, 2012], mutual gaze between interlocutors stands out as the most important clue. [Ferré and Renaudier, 2017] presented the gaze as an important cue for bimodal (visual/vocal and gestural) and visual feedback. Before the production of a visual or bimodal feedback, most of the time, the speaker was looking at the interlocutor.

Summary: Table 1 summarizes the main different feedback-predictive-cues listed in the literature according to their level/modality. We can see a lack of a global model of the feedback predictive cues due to the different phenomena taken into consideration (low pitch regions, final contours, etc.) and it is not directly comparable. However, there were some similar findings, especially prosodic features that seem to play an important role. The contribution as well as the interaction of the relative features are not yet very clear. For example, as shown by [Poppe et al., 2010], performances were reduced when integrating gaze, whereas taking into consideration only pause and pitch strategy gave better results. Conversely, [Truong et al., 2011], did not find significant results for rising and falling pitch, but gaze appeared to be the most relevant indicator.

Cue	Feedback	No Feedback
Prosodic	Flat-fall; rise-fall; high-rise; low-rise High peak of energy Final vocalic lengthening Low/High pitch period Long IPU duration Speech rate Low noise-to-harmonics-ratio	Flat intonation Low peak of energy Short duration of the final phoneme
Lexico-syntactic	POS bigramm: Det-NN; Adj-NN; NN-NN Connective close Disfluencies Final particles	Det Interjections Conjunctions Speech markers
Mimo-gestural	Speaker looks at the interlocutor Nods	Speaker does not look at the interlocutor

Table 1: Summary of feedback and non-feedback predictive cues listed in the state of the art per type of features: Prosodic, Lexico-syntactic and Mimo-gestural.

3. Two Different Ways of Predicting Feedback: Existing Approaches

We can find in the literature two types of method for predicting feedback. The first concerns the temporal prediction and consists in identifying whether or not a feedback may occur, for each timestamp (for example every 50ms). We refer to such a method as “continuous prediction”. The second method consists in studying what happens at specific positions such as pauses. In this case, the task consists in predicting whether the next event after the pause will be a feedback, a turn change or a turn hold. We refer to this task as “non-continuous” prediction.

The works are summarized in Table 2 and 3. They focus on methodological differences in feedback definition, feature selection and evaluation methods.

3.1 Continuous Feedback Prediction

For the prediction of feedback, two main approaches have been explored: the *rule-based* methods and the *machine learning* based approach.

Rule-based methods. The first methods used to predict feedback were *rule-based*. In their seminal paper, [Ward and Tsukahara, 2000] proposed predicting generic-verbal feedback in English and Japanese by means of prosodic features based on five rules presented in Figure 1. This work was based on an audio corpus. For evaluation, the authors looked at the degree of matching between the onset of a predicted feedback and the onset of the observed feedback in their corpus. A window spanning 500ms was used around the onset of the observed feedback. If the time of the predicted feedback belonged to this window, the prediction was considered as correct. The metrics they used are accuracy (defined as the fraction of predictions which were correct) and coverage (the number of items of feedback that the rules could predict). Performances obtained for English had a coverage of 48% and an accuracy of 18%. Japanese showed better results with a coverage of 56% and an accuracy of 34%. From the different algorithms tested, the one based on low-pitch cues provided clearly better results than their baseline (random prediction).

[Cathcart et al., 2003] compared performances of several rule-based models, based on five types of rules on a HCRC MapTask corpus to predict generic feedback. The first baseline model generated

feedback every seven words. The second type of model tested were based on the three or ten most frequent trigram of POS that preceded feedback. The third type of model generated feedback according to different lengths of pause (from 400ms to 1.5s). Finally, [Cathcart et al., 2003] tested the combination of both POS and pauses (10 most frequent trigrams followed by a pause of at least 900ms ; 3 trigrams followed by a pause of at least 600ms). The best model is the one with 3 trigrams and pause duration of 600ms (f-score = 0.35, Recall = 0.43, Precision = 0.29). The window for the evaluation is not specified.

[Poppe et al., 2010] proposed testing 3 rule-based models that jointly and separately used prosodic features (pitch and pause) and visual feature (gaze). They focused only on generic feedback. These models were compared to 3 baselines (copy of human feedback, Ward's rules described above and random generation). The feedback (nod and/or "uh-huh") were produced by a virtual agent. A subjective evaluation was conducted by human observers. Average scores between 0 and 100 were obtained, corresponding to the how natural the feedback produced by the virtual agent was. The prosodic model provided the best result (average score of 43,56%), while considering only gaze gave lower scores (average score of 29,04%). The random strategy produced an average score of 43,39%. Nonetheless, concerning metrics of evaluation used in this paper, the question asked to participants: "*how likely do you think the listener's feedback behavior has been performed by a human listener,*" seems an insufficient evaluation for a complex task such as feedback production. Indeed, the copy strategy gave an average score of 57,67%. Higher results could be expected for a baseline that produces feedback at the same place as the real listener. This score also indicated how difficult it is to evaluate feedback production. We also think that this result may have come from the fact that only nods and "uh-huh" were reproduced, rendering the agent's behavior mechanical. Generic feedback alone is not sufficient to obtain a human-like agent, and support our choice to consider all feedback types.

[Truong et al., 2010] used pitch and pause information to design a rule-based model of feedback. They used the Dutch IFADV corpus, and evaluate their prediction on short vocal feedback (interjections, laughs, short evaluation). This feedback can fulfil the role of generic or specific. A function of a varying evaluation margin is presented from 200ms to 500ms. With a margin of error of 200ms before and after the feedback onset, the f-score obtained is around 0.12. For 500ms, the f-score obtained is around 0.14. As baseline, they compute the rule-based model of [Ward and Tsukahara, 2000]. Results shows that pitch & pause strategy provides a higher precision but a lower recall than the rules of [Ward and Tsukahara, 2000] but, f-score is slightly better with the pitch & pause strategy.

Machine learning methods. Nowadays, the most popular approaches for predictive tasks are based on *machine learning techniques*. [Okato et al., 1996, Morency et al., 2010, Ozkan and Morency, 2012, de Kok et al., 2014, Ozkan and Morency, 2010, De Kok et al., 2010, Huang et al., 2010] employed different algorithms to predict generic feedback. Most of them focused on gestural feedback (nods) [Morency et al., 2010, Ozkan and Morency, 2010, de Kok et al., 2014]. [De Kok et al., 2010, Ruede et al., 2019] also investigated bimodal feedback. A large part of these works made use of the probabilistic sequence model [Morency et al., 2010, De Kok et al., 2010, Ozkan and Morency, 2012]. This method gave an output of a sequence of probabilities. Thanks to these probabilities, a threshold can be used to trigger the prediction (feedback). Thereby, this threshold can be adjusted to match with a level of expressiveness (quantity of feedback produced by the listener). Thus, different types of listening behavior can be modeled. In parallel with the use of more sophisticated algorithms, the models take into account features from different levels of information: prosodic features have been progressively completed with lexico-syntactic, mimo-gestural and temporal cues.

[Morency et al., 2010] used sequential probabilistic models to predict one type of gestural feedback (nods of the head) based on a set of multimodal features (prosody, spoken words and eye gaze). CRF and HMM were used in order to produce distinct peaks of probability across time that could be associated with feedback opportunity. The metrics used to evaluate the models were f-score, precision and recall. Predicted feedback was considered as correct when a peak of probability fell within an interval corresponding to an observed feedback in the corpus (ground truth). The results show an f-score of 0.2562, a precision of 0.1862 and a recall of 0.4106.

In another work, [Ozkan and Morency, 2010] established a new feature-selection method, called *selffeatures*, by looking at the influences of several features individually. Several features were considered from prosodic, lexical, syntactic and visual modalities. Importance of features was computed

for each listener and ultimately a consensus was established on the best features. They finally ran a CRF algorithm to evaluate the nods feedback prediction. An item of feedback was correctly predicted if a peak of prediction fell in the interval of an observed feedback, as in [Morency et al., 2010]. This method provides an improvement for the predictive task. Consensus of self-features made it possible to improve the performances compared to a baseline that included input from all the features. The best scores obtained were an f-score of 0.3196, a precision of 0.23802 and a recall of 0.48629. Different encoding of features were tested (e.g. binary encoding and ramp function encoding that linearly decrease) but we note that noun, determinant, eye gaze and lowness stood out as significant in the improvement of the model. [Ozkan and Morency, 2012] treated other problems regarding prediction in natural conversation, such as the integration of multimodal data features selection and interpretation. In this paper, the authors predicted feedback nods with a new probabilistic model, the *Latent Mixture of Discriminative Experts (LMDE)* that automatically learned a mapping between multimodal observations and a sequence of labels (head-nods). A large set of multimodal features was used (prosodic, visual, morpho-syntactic). Several models were run in order to test the importance of each type of feature (prosodic, visual, lexical, POS and syntactic) and compared with a model that integrated them all. Best results were obtained in the latter case (f-score = 0.3026, precision = 0.1984, recall = 0.6369). Moreover, results of LMDE outperformed other methods tested (early CRF, early LDCRF, mixture CRF).

Deep learning techniques have also been used in different research works. [Mueller et al., 2015] used prosodic features (pitch and power) to predict generic verbal feedback (e.g. “yeah”, “um-hum”, “uh-huh”) on the audio-only Switchboard corpus. Predicted feedback was evaluated as correct if it matched with observed feedback, with a margin of error of 200ms before and after. The resulting f-score was 0.109. In [Ruede et al., 2019], they used a *Long Short-Term Memory (LSTM)* model to predict short forms of verbal feedback. They only predicted when the feedback would be triggered, without making a distinction between the feedback function or type. More than half of the items of feedback were generic interjections (e.g. “yeah”, “um-hum”, “uh-huh”, “right”) but no specification was given regarding the other forms of feedback. We consider, according to their definition of feedback, that at least some of the other forms of feedback could fall under the specific category. Features were extracted automatically (pitch slopes, pause triggers, fundamental frequency variation, Mel-frequency cepstral coefficients, and word history with word2vec) from the audio-only Switchboard corpus. Feedback was evaluated in a given window around the feedback onset. After testing different margins of error, a window of 1s after the feedback onset was chosen. Interestingly, the evaluation was not performed on the entire data but on a sub dataset. Segments where the speaker talked for at least 5 seconds without overlap with the listener since at least 5 seconds are selected. This type of evaluation avoids predictions when the listener holds the floor. Best results show an f-score of 0.388, precision of 0.312 and recall of 0.511.

Metrics for evaluation. Most of the models used classical metrics of machine learning (f-score, precision and recall) to evaluate the performance of the proposed methods. However, as mentioned above, listeners can produce feedback at certain places across plural potential sites of feedback [Howes and Eshghi, 2021]. Not all of these opportunities were chosen. Classical metrics do not measure all of these potential sites of feedback since they used the behavior of one listener as a gold standard. Regarding the evaluation, different methods have been proposed in order to overcome these problems of variability of feedback production. Here we present two new measures for evaluating feedback prediction proposed in [De Kok et al., 2010, Ozkan and Morency, 2012]. The first is called *F-consensus* [De Kok et al., 2010]. This measure considers the feedback produced by three parallel listeners who are all listening to the same speaker (the speaker believes he/she is speaking to only one interlocutor and each listener is not aware of the presence of the other listeners). Precision is calculated on the feedback produced by the three listeners and recall is obtained from the feedback produced by at least two listeners. A consensus is calculated according to a defined window between the sets of feedback produced by different listeners (e.g. if 2 listeners produce a feedback in the same given interval this is a consensus 2, if 3 listeners produce a feedback in the same given interval this is a consensus 3). [De Kok et al., 2010] worked on MultiLIS Dutch corpus (1 speaker, 3 listeners) and used the F-consensus for both training and evaluating their predictive models. Five CRF models were run including 3 with different levels of consensus (consensus 1, 2 and 3) and one baseline, where feedback was produced by only one listener (without consensus). Models were evaluated on different ground truths (only one listener, on consensus 1 and on consensus 2). To sum up, models trained on consensus do not provide

better results than the baseline when evaluation is done on one listener. This result confirmed that features were consistent from one speaker to another and favored the idea that it is only the position chosen by listener to produce it that differs. Indeed, when consensus 1 and consensus 2 are used as ground truths, the baseline (so the feedback displayed by only one listener) provided the worst results (see [De Kok et al., 2010] for complete results). Globally, consensus 2 performs better with an f-score of 0.375 (and f-consensus = 0.439) when the gold standard is consensus 1. Results presented in Table 2 are those of the baseline, since it is the only model that allowed for a comparison with our results. The obtained results finally support the idea that feedback could be successfully predicted at a certain site but considered wrong during evaluation. Finally, this measure is interesting for taking variability into account but this type of data remains difficult to acquire. In addition, the experimental set-up does not allow all types of feedback to be processed, because the speaker is instructed not to formulate explicit answers (excluding a part of specific feedback).

A second metric system called *User-Adaptive Prediction Accuracy* (UPA) was presented by [Ozkan and Morency, 2012]. In this case, the feedback frequency was calculated for each listener and the probability threshold for producing feedback was settled according to this frequency and to the listening behavior adopted by the listener. This metric system aims at capturing the interindividual variability between listeners who produce a lot of feedback and listeners who are more discreet. By comparing results of several models between f-score and with UPA, they found that UPA metrics were more representative than the f-score in finding the best model.

Paper	Method	Feedback	Features	MoE	Scores
[Ward and Tsukahara, 2000]	RB	G-V	P	± 500 ms	P = 0.18 R = 0.48
[Cathcart et al., 2003]	RB	G-V	P M	-	F = 0.35 P = 0.29 R = 0.43
[Truong et al., 2010]	RB	G/S-V	P	± 500 ms	F $\approx$ 0.14 P $\approx$ 0.22 R $\approx$ 0.11
[Ozkan and Morency, 2010]	CRF	G-G	P M G	FB interval	F = 0.32 P = 0.24 R = 0.49
[Morency et al., 2010]	CRF HMM	G-G	P M G	FB interval	F = 0.26 P = 0.19 R = 0.41
[De Kok et al., 2010]	CRF	G-V/G	P M G	-	F = 0.26 P = 0.27 R = 0.26
[Ozkan and Morency, 2012]	LMDE	G-G	P M G	± 500 ms	F = 0.32 P = 0.24 R = 0.49
[Mueller et al., 2015]	DNN	G-V	P	± 200 ms	F = 0.11
[Ruede et al., 2019]	LSTM	G/S-V	P M A	± 1 s	F = 0.39 P = 0.31 R = 0.51

Table 2: Summary of the literature on feedback prediction with objective evaluation. Method column refers to the algorithm: **Rule-based (RB)**, **Conditional Random Fields (CRF)**, **Hidden Markov Model (HMM)**, **Deep Neural Network (DNN)**, **Long Short-Term Memory, Latent Mixture of Discriminative Experts (LSTM)**. The Feedback column refers to the feedback studied, the first letter indicates the type predicted: only **Generic (G)**, or also **Specific (G/S)**; the second letter refers to modality: **Verbal (V)** and/or **Gestural (G)**. The Features column refers to the type of feature: **Prosodic (P)**, **Morpho-syntactic (M)**, **Gestural/Visual (G)**, **Auto-regressive (A)**. **Margin of error (MoE)** column indicates the window used to evaluate feedback from the onset of the ground truth (- indicates missing information). The Scores column contains the metrics and associated scores: **f-score (F)**, **Precision (P)**, **Recall (R)**.

3.2 Non-continuous Feedback Prediction

Current works dealing with interaction modeling are mostly focused on the organization of the interaction and often attempt to predict, individually or simultaneously, either turn-changing, turn-holding or feedback. For a complete review see [Skantze, 2021]. Few works have focused on feedback type classification [Kawahara et al., 2016, Ortega et al., 2020, Jang et al., 2021]. We distinguish these works from those presented above, since the questioning and the methods differ. In this section, we present some works that perform prediction only at a specific moment of the interaction (at the end of an utterance or during the speaker pauses) and do not provided continuous decisions.

[Kitaoka et al., 2006] investigated response timing during non-overlap speech. Silent pauses by the speaker were classified into 3 categories, corresponding to listener behavior: *making feedback*, *turn-taking*, or *waiting*. A C4.5 decision tree was used, based on prosodic, morpho-syntactic features and duration information (e.g. duration of preceding utterance, elapsed time from the end of the previous

utterance). An evaluation with 11-fold validation provided the following scores: f-score = 26.7% for classification in feedback, f-score = 53.9% for turn-taking and f-score = 60.0% for waiting.

[Meena et al., 2014] developed a Response Location Detection system (RLD) in an offline and online fashion from a human-machine corpus. In the training data, a spoken dialog system (the listener) followed the instructions of a human user (the speaker) to perform a MapTask. The Dialog system can produce an acknowledgment, a clarification request, a repetition or a guest response. Each speaker's IPU's were annotated into *Response*, meaning that a response was expected after the given IPU, or *Hold*, meaning that no response was expected. Several models were tested (Naïve Bayes, a J48 decision tree classifier, SVM, Voted Perceptron). The classification between Response and Hold category was performed at the end of each speaker's IPU. The importance of each feature category (prosodic, contextual, syntactic) is tested. Best offline results are obtained with the J48 classifier when all types of features were used. Accuracy was 82.44% on automatic speech transcription and 84.29% on manual transcription. Note that a subjective evaluation was performed with the trained model implemented on the dialog system. Both trained and evaluated models performed better than a majority class baseline.

[Ishii et al., 2021] investigated multitask learning in order to improve feedback predictive models. Multi-modal features (acoustic, linguistic and visual) were extracted to train an Adam optimizer. At the end of every IPU, the model would predict either *feedback*, *turn-changing* or *turn-management willingness* (composed of 4 willingness behaviors: *turn-holding*, *turn-yielding*, *turn-grabbing*, *listening*). A comparison was made between single-task models and multi-task models. Interestingly, the authors proposed to use both speaker and listener signals as features. The results were interesting concerning both the performances and the features given to the model (speaker and listener information). On the one hand, the results decreased when feedback and turn-changing were predicted jointly (the best f-score with both interlocutor features was 0.428). On the other hand, the best performances were obtained when feedback and turn-management willingness were combined, with both interlocutor features (f-score = 0.852). The conclusion was that adding turn-changing to feedback prediction does not improve the prediction performances. On the contrary, predicting both feedback and turn-management willingness allowed an increase in performance. When predicting feedback only, taking both speaker and listener features (f-score = 0.772) outperformed the model considering only the speaker features (f-score = 0.658). While taking just the listener features gives an f-score of 0.752. To our knowledge, this is the first study that demonstrates a significant improvement in performances by adding listener information.

[Kawahara et al., 2016] predicted the morphological form of feedback with Logistic Regression from morpho-syntactic and prosodic features. Short verbal feedback (generic and specific) from a counseling corpus was annotated. The task was to predict the form of the feedback, according to four categories (*un*, *un-un*, *un-un-un* and assessments).² Eight participants discussed their personal troubles with a counselor, in dyadic interaction. Only 2 professionals were engaged to perform the role of a counselor. In order to treat the problem of variability, annotations were augmented. Three human annotators annotated the feedback labels that could be produced in an acceptable way after each IPU or boundary. A label was kept if all of the 3 annotators had selected the same form. The final f-score was 0.656. The model was finally extended in order to predict one more category: *no feedback*. This prediction was made after each IPU and the obtained f-score was 0.643.

More recently, [Jang et al., 2021] proposed a multitask learning task to jointly predict feedback categories (*no-feedback*, *continuers*, *understanding*, *empathic responses*) and a *sentiment score*. Only verbal feedback was considered in this study. The treatment of empathic responses made it possible to process specific feedback. The data used were very specific, since they were from doctor/patient psychiatric counselling sessions. This type of data is ideal for predicting feedback. Indeed, 84% of the doctor utterances were annotated as feedback. The notion of polarity regarding feedback was also considered. As features, prosodic (Mel-frequency Cepstrum Coefficient) and lexical (transformers) information were used. Finally, sentiment score was computed, based on a dictionary containing association between sentiments and word (*strong positive*, *positive*, *neutral*, *negative*, *strong negative*). Using the sentiment feature allowed the authors to outperform previous studies by [Kawahara et al., 2016, Ortega et al., 2020], used as baselines. The authors showed the importance of using lexical

² *un* is a common form used to produce continuers in Japanese.

information that contains polarity in order to improve feedback category prediction, specifically for empathic responses. The f-score reported was 0.76.

Paper	Algorithm	Prediction	Location	Features	Scores
[Kitaoka et al., 2006]	C4.5	FB/TT/W	Pauses	P M	F(FB) = 27 F(TT) = 54 F(W) = 60
[Meena et al., 2014]	J48	Hold/Response	IPUs end	P M C	A = 84
[Kawahara et al., 2016]	Logit	FF	Boundaries end	P M	F = 0.66
[Ishii et al., 2021]	Adam	FB/TT/TMW	IPUs end	A G L	F = 0.85
[Jang et al., 2021]	LSTM KoBert	FT	FB interval	P L S	F = 0.76

Table 3: Summary of the literature on feedback classification in an offline fashion. The Prediction column refers to the classification task: **Feedback (FB)**, **Turn Taking (TT)**, **Turn Taking Willingness (TTW)**, **Waiting (W)**, **Feedback Form (FF)**, **Feedback Type (FT)**. The Location column describes the site where classification is performed. The Features column refers to the type of features used: **Prosodic (P)**, **Morpho-syntactic (M)**, **Gestural (G)**, **Contextual (C)**, **Lexical (L)**, **Sentiment (S)**, **Acoustic (A)**. The Scores column presents the metrics used and scores: **Accuracy (A)**, **F-score (F)**.

As shown in this section, we have presented some works that focus on feedback prediction, either to predict potential sites of feedback (continuous prediction), or to predict feedback types. The difficulties highlighted for an objective evaluation and the large variability in the listening behaviors, explain why the obtained f-score in the literature for continuous prediction never exceeds 0.40. Moreover, this score also depends on the window of evaluation (margin of error). Scores for prediction that focus on specific moments (end of IPUs, pauses, etc.) of the interaction or the type of feedback are pretty good, but predicting the type of feedback still needs to be investigated.

Based on this substantial review, we now aim at presenting data, methodological choices and models performed in order to improve the prediction of all feedback in conversation. The first model concerns the level of prediction of feedback position and the second the level of prediction of feedback type.

4. Dataset

In this section, we present our typology of feedback and the PACO-Cheese! corpus used in our study. Next, we will provide some description of the data. Features annotation and extraction are described in the final section.

4.1 Feedback Typology

The feedback considered was multimodal. We postulate here that all feedback whatever its modality can be classified into classes and sub-classes of function (understanding, agreement, disagreement, acknowledgment, assessment and cohesion).

We proposed a hierarchical classification based on only two classes (**generic** and **specific**) and two sub-classes for the specific feedback (**positive/negative**; **expected/unexpected**). This enabled us to take into account the multi-functionality of the said feedback (see also for a previous work [Boudin et al., 2021]).

Generic vs. Specific: Generic feedback is a consistent phenomenon which mainly takes the form of a nod, an interjection, a smile or a combination/repetition of these elements. The functions of generic feedback are more restricted than those of specific feedback and simply show comprehension and agreement. Since generic feedback is homogeneous in form (limited to a defined list of realizations [Prévo et al., 2016]) and in function, they do not need a more detailed typology. Conversely, specific feedback is more context-dependent and related to semantic interpretation. It can be composed of various visual and/or vocal elements (marked intonation, longer lexicalization, laughter, eyebrow movements, smiles, head movements or facial expressions). Specific feedback can convey different

attitudinal/emotional values that we represent by a finer classification using two levels of sub-classes, illustrated in Figure 4.1.

Positive vs. Negative: Firstly, we evaluated feedback regarding to what the interlocutor was reacting to. *Did the feedback react to something positive or to something negative? Did the speaker talk about an experience or an event that he/she was evaluating negatively by expressing a criticism, sadness or hunger, etc? Or on the contrary, was he/she evaluating positively with joy or humor, etc?* Note that we did not consider negative feedback as feedback that was incorrectly produced/showed disagreement/was rejected by the main speaker, as other works have done in the past. This positive versus negative aspect of feedback represents the polarity associated to the semantic content produced by the main speaker and to which the listener responds. It reflects the polarity alignment between participants.

Expected vs. Unexpected: The second level of distinction concerns the *expected* or *unexpected* aspect of the information to which the listener is reacting [Prévot et al., 2017]. The question is whether the listener expects to receive this information at this moment, whether he/she already knows this information, and whether the production of this information at that moment of the interaction was expected. This level of feedback definition allowed us to distinguish feedback that showed new instances in the common ground rather than that which reacted to a pre-existing common ground.

We propose this classification in order to avoid ad hoc and corpus-dependent representations such as those often used in other works. Figure 2 illustrates the proposed specific feedback classification and provides examples of associated attitudes.

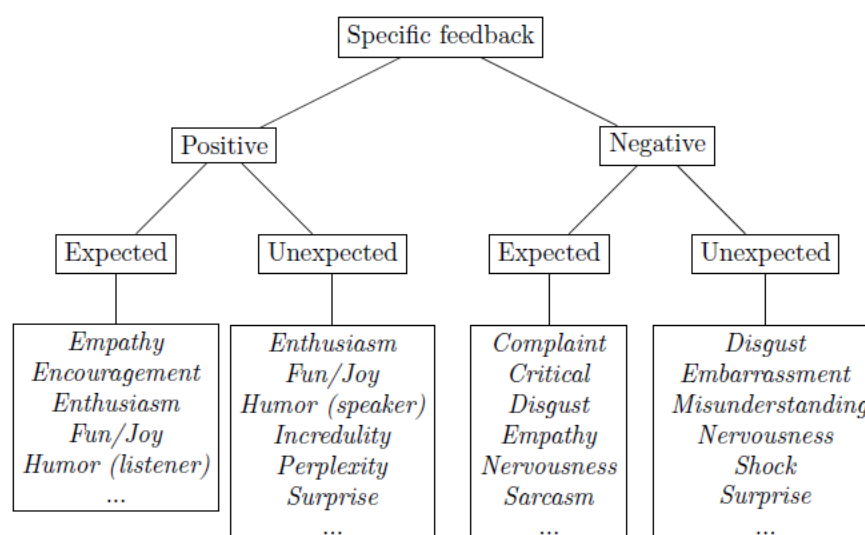


Figure 2: Specific feedback classification scheme. The 1st level contains the classification of Positive/Negative feedback. The 2nd level contains Expected/Unexpected feedback. The 3rd level contains some examples of attitudes per specific feedback types.

4.2 PACO-Cheese! Corpus

Our dataset is built upon two corpora of natural conversations in French: [dataset] PACO [Amoyal et al., 2020] (<https://www.ortolang.fr/market/corpora/paco>) and Cheese! [Priego-Valverde et al., 2020] (<https://www.ortolang.fr/market/corpora/cheese>), referred as PACO-Cheese!. It contains 7 hours of audio-visual recordings. Participants (dyads) were installed face-to-face. They first had to read a short story before having a free conversation. PACO-Cheese! is composed of 26 interactions, each lasting between 15 and 20 minutes. The participants of Cheese! (about half the dyads) already knew each other beforehand whereas the participants in PACO (the other half) were meeting for the first time. It is enriched with different annotations. A manual transcription was performed, including varied information such as noise, laughter, pauses, elision and unexpected events. This transcription was automatically aligned with the signal thanks to the SPPAS system [Bigi, 2012, Bigi, 2015] (<http://www.sppas.org/>) which returns the list of phonemes, syllables and IPUS. The MarsaTag analyzer

[Rauzy et al., 2014] (<https://www.ortolang.fr/market/tools/sldr000841/v1#!>) was then applied to extract lemmas and POS. Moreover, smiles and nods were annotated semi-automatically thanks to SMAD (<https://github.com/srauzy/HMAD>) [Rauzy and Amoyal, 2020, Amoyal and Priego-Valverde, 2019, Boudin et al., 2021].

This corpus provides valuable verbal and gestural information for the development of a predictive feedback model. We further enriched the existing annotations at different levels, in particular in terms of prosody and gestures, by means of different tools [Hirst, 2007, Hirst, 2022, Rauzy et al., 2014, Rauzy and Amoyal, 2020, Bigi, 2015, Baltrusaitis et al., 2018].

Many works have studied the importance of particular gestures such as nods [Schegloff, 1982, Allwood and Cerrato, 2003, Stivers, 2008]. They are frequently used during production and represent the most common gestural feedback. Nods are vertical movements of the head, that can be realized in a single form (one movement from the bottom to the top or the other way around) or in a plural form (sequence of several nods). They can be used by a speaker to underline an important part of the speech or by the interlocutor as feedback. Feedback nods are generally used as generic feedback, but can also be specific when they complete certain verbal or facial expressions [Allwood and Cerrato, 2003]. They can be produced in a unimodal (nod alone) or in bimodal way (e.g. nod + yes). Nod production was annotated in a semi-automatic way: we first applied an automatic extraction of nod productions and then did a manual correction. The automatic annotation was performed with the HMAD open source tool [Rauzy and Goujon, 2018]. Technically, detecting nods consists in looking for sinusoidal movement of the head and returning the temporal interval during which the movement is performed. HMAD takes as input a front view video treated by the OpenFace software [Baltrusaitis et al., 2018] and outputs the annotation in an ELAN format. The following step consists in correcting the ELAN output manually [Sloetjes and Wittenburg, 2008]. Smiles were annotated in a similar procedure as described in [Amoyal and Priego-Valverde, 2019]. They were annotated on 4 levels (S1, S2, S3 and S4) and neutral faces were annotated with the S0 label as proposed in [Rauzy and Amoyal, 2020].

Finally, feedback was annotated on 13 dyads, according to our typology (Figure 2) which represents approximately 3.6 hours of recording. The current work is based solely on this part of the corpus. A total of 2,377 items of feedback were obtained (1,206 generic and 1,171 specific, including 416 positive-expected, 548 positive-unexpected, 115 negative-expected and 92 negative-unexpected).

4.3 Description of Data

In this section we describe the distribution of feedback over 13 dyads of the PACO-Cheese! corpus according to our typology. We also propose to look at the main elements that composed the feedback forms. The frequency of the feedback observed in our dataset firstly confirms the literature [Pope et al., 2010]: feedback is frequent and its production is a consistent phenomenon. Our data show a frequency of 10.78 items of feedback per minute (see Table 4), the standard deviation of this frequency being quite high ($\sigma = 3.02$). The less expressive interlocutors had a tendency to produce half as many items of feedback compared to the more expressive interlocutors: this observation also confirms the problem of individual variability. The mean duration of feedback is 1.27s (minimum duration = 400ms, maximum duration = 6.52s).

Feedback type		Frequency per minute
Generic		5.56 (± 2)
Specific	Positive-Expected	1.88 (± 1)
	Positive-Unexpected	2.44 (± 1.58)
	Negative-Expected	0.58 (± 0.43)
	Negative-Unexpected	0.44 (± 0.31)
	Total Specific	5.14 (± 2.58)
Total Feedback		10.78 (± 3.02)

Table 4: Feedback frequency per minute and per type: Generic, Positive-Expected, Positive-Unexpected, Negative-Expected, Negative-Unexpected, All specific Feedback and All Feedback.

Distribution between generic and specific feedback is roughly equivalent. Regarding the sub-types of specific feedback, we note they are more often positive than negative. Overall, the most frequent type of feedback was the positive-unexpected type, the least common was the negative-unexpected type. Table 5 shows the total number and the frequency per minute of IPU, tokens, nods, laughs and smiles.

Item	Total	Frequency per minute	Mean dur. (s)	Min. dur. (s)	Max. dur. (s)
IPU	7505	17.25 ($\pm$ 5.68)	2.51	0.10	19.43
Token	50 303	115.83 ($\pm$ 36.56)	0.49	0.015	2.74
Nod	3699	8.52 ($\pm$ 3.65)	0.99	0.10	5.79
Laugh	1063	2.45 ($\pm$ 1.95)	0.95	0.04	5.48
Smile 3 and 4	1540	3.55 ($\pm$ 1.67)	3.72	0.2	17.26

Table 5: Frequency per minute, total count, mean duration (Mean dur.), minimum duration (Min. dur.) and maximum duration (Max. dur.) of: IPU, Token, Nod, Laugh and Smile of S3/S4 intensity on the 13 dyads of PACO and Cheese!.

It is interesting to note how nods, laughs, verbalization and smiles are used according to the generic/specific function of the feedback. Figure 4 indicates the amount of feedback produced with at least one of these items (among a total of 2,377). Figure 5 presents the quantity of generic and specific feedback produced in the three modalities studied: bimodal, unimodal-visual and unimodal-vocal. Generic and specific feedback is mainly produced in verbal forms (interjections, repetitions and other lexical elements). Specific feedback being context dependent, is produced using more specific lexical elements whereas generic feedback generally uses short items like interjection. As expected, nods are significantly more frequent for generic than specific feedback, confirming [Bavelas et al., 2000]. Conversely, laughter and high intensity smiles (S3, S4) are widely used as specific feedback and are quite rare in generic feedback. Finally, generic feedback is mainly produced with nods and/or verbalization, whereas specific feedback is primarily composed of verbalization/laughter and/or smiling.

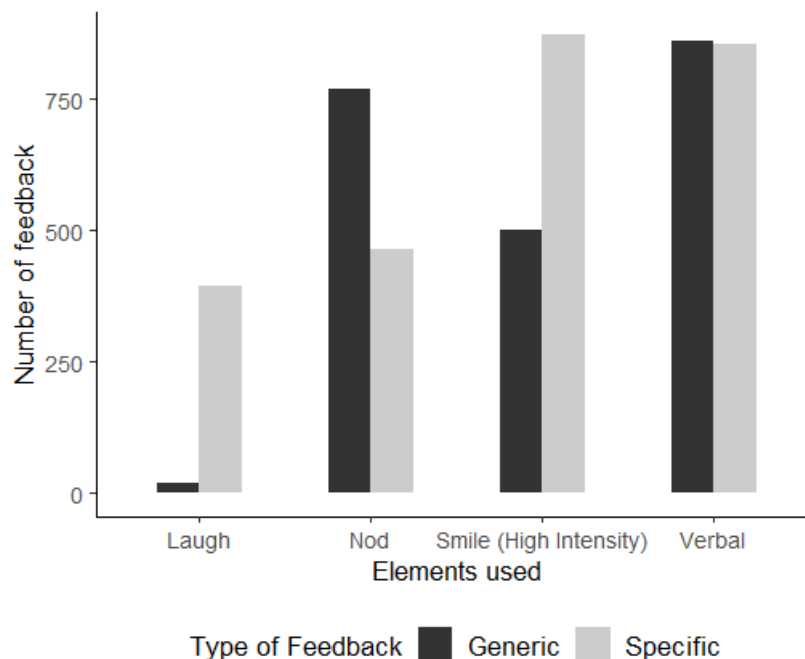


Figure 4: Items used to produce generic and specific feedback. Numbers of Generic and Specific types of feedback produced with at least a Laugh, a Nod, a High Intensity Smile (S3, S4) or a Verbalization.

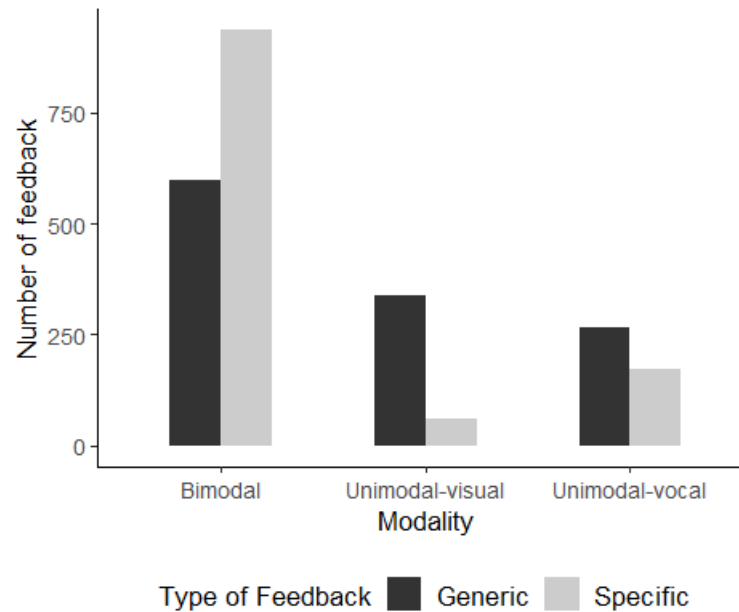


Figure 5: Modality used to produce feedback: Bimodal; Unimodal-Visual; Unimodal-Vocal, according to their generic/specific type.

We noticed that 47% of the feedback was produced in verbal overlap. By verbal overlap, we mean that both speaker and listener were producing a verbal signal at the same time. There was an equal amount of generic and specific feedback produced with a verbal overlap. If we look at all the feedback, 76% was produced during the speaker speech, either partially or totally. To better understand this result, we also looked more precisely at how this overlap occurred. We observed that 31% of feedback (all types and modalities considered) was totally produced during the speaker speech. We noticed that surprisingly 53% of unimodal-vocal types of feedback (verbalization and laughter) was produced in a total overlap, which indicates that vocal feedback is not preferentially used during pauses. We also notice that only 36% of vocal feedback was totally produced during pauses. Nonetheless, if we look all the feedback (all types and modalities considered), only 20% of it was produced during pauses only. Finally, half of the feedback was produced during moments where the speaker was switching between speech and pauses. 15% of feedback started during speech and ended during a pause. 17% of feedback started during a pause and ended in overlap. Finally, 18% of feedback was produced either in a pause/overlap/pause or overlap/pause/overlap pattern. These observations confirm our claim that studying feedback only during the silent pauses of the main speaker, considerably reduces the quantity of feedback.

4.4 Features Extraction

In this section, we present the different features we used and how we extracted them according to the first level (feedback position) and the second level (feedback type) of prediction, presented in Sections 5 and 6.

4.4.1 Prosodic features

Concerning prosody, we integrated the intonation patterns given by a sequence of tones (on top of the pauses). Tones represent an intermediate level between low-level acoustic features (e.g. pitch) and phonological interpretations. In some cases, at the end of a sequence potentially bounded by a pause, tone patterns corresponded to a final intonation contour. The intonation contour could be correlated with the introduction of new information and possibly carried an important part of the interactional meaning. Several studies have also shown that this final contour might be a good predictor for feedback occurrences. Taking into account tone patterns (encoded in our approach by n-grams) makes it possible to compare the influence of tones taken separately or by sequence. Tone extraction is done automatically, thanks to the pitch modeling tool MOMEL-INTSINT [Hirst, 2022]. This processing relies

on two steps. The first consists in modeling the f0 based on a sequence of transitions between successive points on the curve (called anchor points). This step (corresponding to the calculation of MOMEL) is based on the relationship between the median, minimum and maximum values of each speaker's pitch range. The *Octave-Median Scale* makes it possible to compare speakers with different pitch ranges (for example males versus females). In a second step, the MOMEL anchor points are automatically encoded into an alphabet of tonal symbols *T(op)*, *B(ottom)*, *M(id)* referring to absolute values and *H(igher)*, *L(ower)*, *S(ame)*, *U(pstepped)*, *D(ownstepped)* referring relative values. This encoding provides intonation patterns represented by the key/midpoint and the span of the speaker's pitch range. To extract the features from the tone annotation, for the 1st level of prediction, we extracted before each timestamp of 40ms the 3 last tones produced. Next we extracted only n-grams of tones that were produced more than a given occurrences threshold to limit the number of features. We used two thresholds of occurrence (500 and 800). If an n-gram was present more than the given threshold it was then binary encoded for each timestamp as a feature³. For the 2nd level of classification, we also tested both a window of extraction of 2s and the 3 last n-grams. The thresholds of occurrence used were 10, 15, 20, 50, 100. Prosodic models considered silent pauses to be a boundary cue of the intonational phrase. An intonation contour associated with a pause could reinforce the end of such a major prosodic unit that could be associated with the end of a discursive unit and that promote feedback. We also encoded the duration of the silent pause (200-400ms, 400-600ms, 600-1200ms, 1200ms and more). Speech rate was also considered (the number of tokens produced by the main speaker in the two previous seconds). Finally, we also automatically annotated when the main speaker was currently speaking for the 2nd level of prediction, feature that we called *overlap*. This feature indicated if the main speaker was speaking when feedback occurred.

Level of prediction	Features	Encoding	Extraction window(s)	Threshold of occurrence
1st	Number of tokens	Count	2s	500, 800
	N-grams of tones	Binary	3 last	-
	Pauses 200-400ms		LB	
	Pauses 400-600ms			
	Pause			
	Pauses 600-1200ms			
	Pauses 1200ms-			
2nd	Number of tokens	Count	2s	-
	N-grams of tones	Binary	3 last ; 2s	0, 20, 40, 50, 1000
	Overlap		FB	-
	Pauses 200-400ms		LB	
	Pauses 400-600ms			
	Pauses 600-1200ms			
	Pauses 1200ms-			

Table 6: Summary of Prosodic features and their extraction according to the task performed: 1st level of prediction and 2nd level of prediction. The table presents the **feature name**, its **encoding**, its **window of extraction** before the timestamp (**3 last elements (3 last)**, **overlap with the left border of the timestamp (LB)** and **2 seconds before (2s)**), and the **threshold of occurrence for n-grams**.

4.4.2 Morpho-syntactic features

On the lexico-syntactic level, we proposed to include POS as well as lexical semantic information. POS are often involved in the literature for predicting feedback. Some of them (for example adverbs, [Bertrand et al., 2007]) have been shown to play a role in favoring feedback. POS can also provide important information at the discourse level: for example, discourse markers reveal the discourse structure, and are often associated with transitions between discourse units (that may correspond to

³ Note that the count is realized on the sliding window data-frame, then when a n-gram is produced, it is annotated several times.

listener's reactions). We proposed then to involve them in the model sequences of POS encoded with n-grams. Note that in our POS tagging, punctuation was also taken into account, although more often used for written text. This punctuation helped us to segment speech into units of varying degrees of completion because they were likely to be associated with the end of a sentence (strong) or with the end of a clause/syntagm (smooth). All POS and punctuation tags were automatically extracted with MarsaTag [Rauzy et al., 2014]. POS features were extracted and encoded as described above in the same way as the tones n-gram. Punctuation was extracted in a previous window of 2 seconds before each timestamp.

Besides POS, we extracted lexico-semantic information about word polarity (positive, negative) and aspect (concreteness) on the basis of word lists given in [Bonin et al., 2018]. These features were extracted in a window of 2s before each timestamp. They were binary encoded (presence or absence) for the 1st level of prediction. For the 2nd level of prediction, they were counted since the last feedback produced, resulting in a numerical encoding. This information is important in particular when studying specific listener reaction (emotion, surprise, introduction of new discourse referents, etc.).

Level of prediction	Features	Encoding	Extraction window(s)	Threshold occurrences
1st	N-gram of POS	Binary	3 last	500-800
	Positive words		2s	-
	Negative words			
	Concrete words			
	Discourse markers			
	Punctuation			
2nd	N-grams of POS	Binary	3 last ; 2s	10, 20, 40, 50, 1000
	Positive words	Count	Last FB	-
	Negative words			
	Concrete words			
	Discourse markers			
	Punctuation	Binary	2s	

Table 7: Summary of the Morpho-syntactic features and their extraction according to the task performed: 1st level of prediction and 2nd level of prediction. The table presents the **feature name**, its **encoding**, its **window of extraction** before the timestamp (**3 last elements (3last)**, **2 seconds before (2s)**, **time since last feedback (Last FB)**), and the **threshold of occurrence for n-grams**.

4.4.3 Mimo-gestural features

The introduction of gestures as a feature completed the multimodal description: nods, laughs and high intensity smiles (S3 and S4, referred to as *smiles*) were taken into account in the prediction. Their extraction was previously described in section 4.2.

For the 1st level of prediction, two windows were tested to extract nods, laughs and smiles. For the first window, multimodal features were extracted in the previous timestamp of 40ms. For the second window, they were extracted in a previous window of 2s. Regarding the 2nd level of prediction, mimo-gestural features were extracted in a previous window of 2 seconds.

Level of prediction	Features	Encoding	Extraction window(s)
1 st	Nods	Binary	LB : 2s
	Smile		
	Laughs		
2nd	Nods		2s
	Smiles		
	Laughs		

Table 8: Summary of Mimo-gestural features and their extraction according to the task performed: 1st level of prediction and 2nd level of prediction. The table presents the **feature name**,

its **encoding**, its **window of extraction** before the timestamp (**overlap with the left border of the timestamp (LB)** and **2 seconds before (2s)**).

4.4.4 Auto-regressive features

As well as the speaker production context, it is necessary to consider what was produced previously by the listener: the prediction of feedback also depends on the previous feedback occurrences. We proposed to integrate this information into the model by means of such “*auto-regressive*” features implementing the fact that the source of information does not come only from the speaker but also from the listener. A system (in our case a participant of the interaction) is able to self-regulate according to his/her memory of previous actions.

Concretely, auto-regressive features encode information about the amount of time elapsed since the last feedback has been produced. In order to produce binary features bearing precise information, we encoded the information into 5 classes, depending on the duration since the last feedback (0-2 seconds, 2-5 seconds, 5-10 seconds, 10-20 seconds, 20 seconds and more since last feedback).

Finally, we also added a categorical feature indicating the type of the previous feedback produced for the 2nd level of prediction (0 = generic ; 1 = positive-expected ; 2 = positive-unexpected ; 3 = negative-expected ; 4 = negative-unexpected).

We would like to clarify one point concerning the source of this type of information. For an online prediction engine, auto-regressive features must be computed dynamically based on the previous predictions already made by the engine. However, in our case, during the evaluation step, our predictive model worked in an offline mode. This meant in particular that all events were already established. We knew that the feedback produced guided and supported the course of the interaction and could simultaneously modify the signal of the speaker and the feedback inviting-cues produced, that were processed by the listener. At this stage of our work, we evaluate the predicted feedback based on the observed feedback, anchored in an unchangeable context. Thus, for this work, auto-regressive features were computed concerning the observed feedback, instead of the predicted feedback.

Level of prediction	Features	Encoding	Extraction window(s)
1st	Last FB 0-2s	Binary	2s
	Last Fb 2-5s		5s
	Last FB 5-10s		10s
	Last FB 10-20s		20s
	Last FB 20s		- s
2nd	Last FB type	Discrete	Last FB

Table 9: Summary of Auto-regressive features and their extraction according to the task performed: 1st level of prediction and 2nd level of prediction. The table presents the **feature name**, its **encoding**, its **window of extraction** before the timestamp (**2/5/10/20 seconds before and time since Last Feedback (Last FB)**).

5. First Level of Prediction: Predicting Feedback Positions

This section presents the methodology implemented for the continuous prediction of feedback (every 40ms) and more precisely data sampling, data pre-processing, features selection, construction of the model, results performances and discussion.

5.1 Data sampling

In order to predict the precise locations of feedback occurrence we defined two timescales. The first one was the sampling rate, the second was the timescale of the extracted feature.

This question of temporal segmentation is addressed quite differently from one study to another. The main difficulty is that predicting feedback location requires us to examine all possible positions, which means applying high frequency sampling. As a consequence, the set of positions where no feedback occurs is by far the largest. In a previous work by [Boudin et al., 2021] it was proposed that all time locations corresponding to the end of verbal or non-verbal units be considered (e.g. end of a word, pauses, laughter, smiling, etc.). Each observation generated an entry of the data table. The value of the

dependent variable was set at 0 for encoding absence of feedback and at 1 for observed feedback. The task therefore uses a standard binary classification. In our case, the main difficulty comes from the unbalanced class distribution: “*no feedback*” events are by far the most frequent. The problem in this case is that a model predicting only no-feedback, whatever the position, would still lead to good accuracy.

Hereafter in the current study, we chose a sampling rate of 40 milliseconds (i.e. a time step) which was in accordance with the frame rate of the captured video. This sampling frequency made it possible to register rapid changes inherent to the interaction and determined the accuracy within which the feedback onset was predicted. A second timescale specified the size of the time window from which the features were extracted. This timescale duration in practice depends on the features considered (as describe in Tables 6, 7, 8 and 9). This in mind, our input data was sampled at increasing timestamp (i.e. the series of time incremented by the given time step). For each timestamp, the presence of the observed feedback received Boolean encoding, i.e. 0 before the feedback location and 1 from the feedback onset until the end of the feedback.

5.2 Preprocessing

The model uses each interaction from the corpus as input twice. Once with interlocutor A considered as the speaker (features were therefore extracted from his/her signal) and interlocutor B as the listener (the one who produced feedback, the binary dependent variable). The same interaction was then used a second time while reversing interlocutor roles. A preprocessing to discriminate the role of speaker and listener is necessary: without it, the features from the complete signal produced by an interlocutor would be processed (regardless of whether it is in speaker or listener position). This would introduce a bias in the learning process and at the same time some features selected by the model would indicate a listener position (e.g. interjections such as “*mmh*”, “*okay*”, etc.), and would therefore favor “non-feedback” prediction. [Ruede et al., 2019] tried to overcome this problem by only selecting the periods where the speaker speaks for at least 5 seconds and the listener has been silent for at least 5 seconds. In our research work, we only considered frames during which the speaker had produced more tokens than the listener in the 2 previous seconds (unless the listener produced feedback). As a result, we mainly kept frames in which the speaker holds the floor and we considered all the feedback, whether it was produced in overlap or not. The dataset size was then drastically reduced to strictly relevant information: the initial dataset contains 660,334 frames (each 40ms), and 326,674 frames after this pre-processing. This method also allowed us to reduce the problem of unbalanced classes by reducing the number of non-feedback frames.

5.3 Feature engineering

Different combinations of features and extraction of features have been tested in the presented work. N-grams of tones and of POS represent an important part of our features. They were selected according to their distribution in the dataset. Their occurrence was calculated on a sliding window containing the three last POS and the three last tones produced by the speaker before the end of the timestamp. The n-gram was included in the features set when it occurred more than a given threshold in the dataset, as indicated in section 4.4.

Several tests were set for feature extraction in order to represent information in the most efficient way and to reduce the multidimensionality of the data. Firstly, we tried different selection thresholds (500, 800) for POS and tones n-gram selection. Secondly, we reduced the number of features by clustering them. The different bigrams and trigrams of tones were clustered into intonational patterns (*falling*, *rising*, *flat* intonation and their combinations e.g. *rising-falling-rising*). On the basis of the pitch range calculated by MOMEL-INTSINT, we encoded the span with two additional features: *small* or *large* span. Finally, besides using detailed morpho-syntactic information (i.e. the POS category plus its features such as *Ppd* encoding a dative personal pronoun), we also simply used the POS label (referred below as meta-POS vs. detailed-POS). Meta-POS were selected with a frequency threshold of 500 occurrences.

We tested several feature combinations, based on these different types of encoding, for POS and tones, without modifying other feature encoding:

- Meta-POS & tones 800 (total of 420 features)

- Meta-POS & tone patterns (total of 273 features)
- Detailed-POS 500 & tones 500 (total of 524 features)
- Detailed-POS 800 & tone patterns (total of 255 features)
- Detailed-POS 800 & tones 800 (total of 402 features)

We also tested the performances of the model with mimo-gestural features extracted immediately preceding each timestamp (with an overlap in the left border of the timestamp), and when they were extracted in a previous 2-second window.

	500	800	Clustered
POS	272	214	232
Tones	232	168	21

Table 9: Number of features given as input to the model, with all features described in section 4.4 and 3 different levels of information for POS and tones. 1st column used n-gram of POS and of tones when they are extracted with a threshold of occurrences of 500. 2nd column used n-grams of POS and of tones when they are extracted with a threshold of occurrences of 800. 3rd column used intonational pattern of tones and main POS categories (metaPOS). Detailed are given in 5.3.

5.4 Building the predictive model

We used Logistic Regression (*Logit*) to build our model firstly because it provides the probability of feedback realization instead of a simple classification. Note that this characteristic is interesting in the perspective of implementing the model in a human-machine communication system, making it possible to introduce a fine variability in feedback production. Secondly, using *Logit* also provides the possibility of evaluating feature importance, which is essential for interpretation of the results. Finally, *Logit* is also appropriate when dealing with small datasets, being less prone to overfitting. Classically, the probability of producing a given type of feedback (or the probability that the feedback is produced at a given time location) is modeled by the equation:

$$\text{logit}(p) = \ln\left(\frac{p}{1-p}\right) = a_0 + a_1x_1(t) + \dots + a_jx_j(t) + \dots \quad (1)$$

where $x_j(t)$ are the predictors which depend on the time location t , and can adopt binary, categorical or continuous types. In a first step, the parameters of the *Logit* model (i.e. the a_i coefficients) are estimated based on the training sample. An analysis of the result makes it possible to decide which predictor contributes significantly to the prediction. The model is finally built with the set of relevant predictors and a probability p is attributed to any combined values of the predictors.

Concretely, the position of the feedback produced by a listener is predicted as follows. The system is a two states automaton, the first state q_0 corresponds to non-feedback positions, the second state q_1 to places where feedback can occur. The system is initialized at time $t_0 = 0$ on the q_0 state and evolves at each time step Δt following the probability given by the *Logit* model, i.e. the probability $p(t)$ that the feedback onset occurs within the time interval $[t, t + \Delta t]$. This probability depends on the values of the features at time t : the features characterizing the production of the speaker and the auto-regressive features of the listener which specify in particular the current state of the system (q_0 or q_1) but also some other features such as the time elapsed since the production of the last listener feedback. As the system evolves, we compute the cumulative probability that the system remains in the q_0 state. This probability is given by the product of the individual model probability starting at the time the last feedback was produced, $P_i(q_0) = \prod (1 - p(t_i))$. Once the cumulative probability falls below 1/2, feedback is generated. The procedure is illustrated Figure 6. Other methods which associate the positions of the feedback with the local maxima of the probability curve can be envisaged as well [Morency et al., 2010].

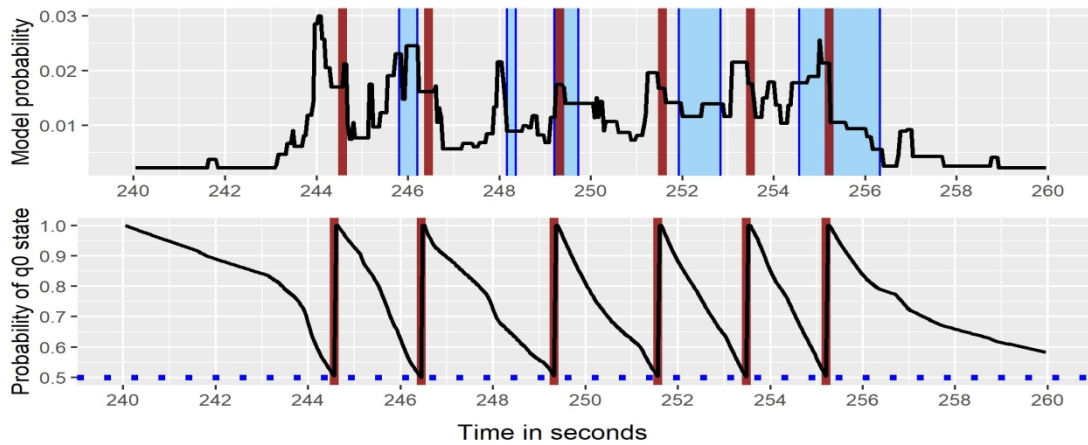


Figure 6: Illustration of the method to predict feedback locations. From the measurements along the time line of the selected features, the model predicts the variation of probability that feedback occurs (the black curve on the top panel). The cumulative probability of the system to remain in the q_0 non-feedback state (the black curve on the bottom panel) is computed. Once this probability falls below 1/2 feedback is generated (the vertical red lines) and the cumulative probability is reinitialized for the next prediction. Predicted feedback locations are close to maxima of the Logit model probability. Blue areas identify the true locations of observed feedback for this slice of interaction. [color should be used]

In order to obtain relevant features and their importance, we first trained a global model with all the features for the five combinations described in Section 5.3. For each feature the model returned an associated coefficient as presented in equation 1. The amplitude of the coefficient traces the strength of the contribution for the given feature. A positive coefficient indicates that the presence of the feature enhances the probability for the listener to produce a feedback whereas a negative coefficient decreases this feedback probability. The logit model also provides the confidence level interval associated with each coefficient estimate. A feature is selected if its estimate is significantly different from 0 within the confidence error bars. In practice this makes it possible to discard features which are not relevant for feedback prediction.

Finally, cross-validation is applied by running a Monte Carlo cross-validation (on 50 trials with a ratio 80%-20% for the training versus the evaluation sample). This evaluation procedure is also applied in order to measure the relative contribution of each feature combination (e.g. grouped by modalities).

5.5 Results & Discussion

The best model was obtained by using the combination of POS-500 and tones-500, with mimo-gestural features extracted just before the timestamp⁴. For the target prediction (feedback), features from each subset of features were selected. The model finally consisted in 112 significant features, including 53 that promoted *feedback* and 59 that promoted *non-feedback*. Feature coefficients are presented in Figure B (see Appendix B) ranked by the importance of their contribution to the model. The strength of the contribution for each feature is traced herein by the amplitude of the coefficient estimate.

Concerning those features which discourage feedback production, only lexico-syntactic and prosodic features stand out as being significant. Table 10 presents the results (f-score, precision and recall) obtained with our model. We compared these results with two baselines. The “*Regular*” baseline generates a feedback every 5.56s. This value corresponds to the frequency of feedback observed in our data (10.78 items of feedback per minute). The “*Random*” baseline generates a certain number of feedback items based on the mean frequency per minute but at random places.

We evaluated our model according to three different windows of ground truth for comparison with the literature. The first, applied a margin error (“*MoE around the feedback onset*” in the table) of 500ms before and after the onset of the feedback, as proposed in [Ward and Tsukahara, 2000]. In other words,

⁴ Note that differences in results are not of deep importance between the two windows of extraction for the mimo-gestural features

feedback is correctly predicted if its onset falls between 500ms before and 500ms after the onset time of the observed feedback. The f-score obtained in this case was 0.28. The same window type, but with a larger margin of error (1000ms) was used, leading to an f-score of 0.49. The second window (“Feedback timestamp as MoE” in the table) is equivalent to that used in [Morency et al., 2010], when the ground truth was the time-span of the observed feedback. In this case, feedback was correctly predicted if its onset time fell between the start time and the stop time of the observed feedback. The f-score value was 0.37.

Feedback prediction	MoE around the feedback onset				Feedback timestamp as MoE		
	Window	F-score	Precision	Recall	F-score	Precision	Recall
Logit	500ms	0.28 (± 0.018)	0.23 (± 0.016)	0.34 (± 0.022)	0.37 (± 0.019)	0.31 (± 0.019)	0.46 (± 0.022)
	1s	0.49 (± 0.022)	0.40 (± 0.021)	0.63 (± 0.027)			
Regular baseline	500ms	0.21 (± 0.016)	0.16 (± 0.013)	0.31 (± 0.026)	0.24 (± 0.019)	0.18 (± 0.018)	0.37 (± 0.030)
	1s	0.38 (± 0.018)	0.28 (± 0.015)	0.58 (± 0.029)			
Random baseline	500ms	0.19 (± 0.013)	0.15 (± 0.012)	0.26 (± 0.020)	0.19 (± 0.014)	0.14 (± 0.012)	0.29 (± 0.020)
	1s	0.34 (± 0.0018)	0.28 (± 0.0018)	0.43 (± 0.026)			

Table 10: F-score, Precision and Recall for potential feedback sites with *Logit*. The regular baseline predicts feedback every 5s, based on the frequency of feedback observed in the data. The Random Baseline predicts feedback at random places by respecting the proportion of feedback/non-feedback instances observed in the data. Results of the same models are presented with 3 margins of errors (MoE): ± 500 ms as in [Ward and Tsukahara, 2000] and ± 1000 ms around the feedback onset and timestamp of observed feedback as in [Morency et al., 2010].

Table 11 compares results of the model trained and tested with all the features vs. models trained and tested with only one type of feature, using an evaluation window with a margin of error at 500ms. As anticipated in our hypothesis, the best performance was reached when using all features (f-score = 0.28). In terms of relative contribution (without interaction), auto-regressive features lead to an f-score of 0.24, with morpho-syntactic features having an f-score of 0.21, prosodic features an f-score of 0.19 and finally mimo-gestural features an f-score of 0.18. Regarding the importance of auto-regressive features, we also tested our model with only features from the speaker (prosodic, morphosyntactic and mimo-gestural), giving an f-score of 0.22. Auto-regressive features are thus important but need to be combined with other types of information.

Modality	Window of 500 ms		
	F-score	Precision	Recall
All	0.28 (± 0.016)	0.23 (± 0.015)	0.34 (± 0.019)
Auto-regressive	0.24 (± 0.016)	0.20 (± 0.016)	0.30 (± 0.017)
Morpho-syntactic	0.21 (± 0.015)	0.18 (± 0.02)	0.24 (± 0.017)
Prosodic	0.19 (± 0.017)	0.16 (± 0.016)	0.22 (± 0.017)
Mimo-gestural	0.18 (± 0.02)	0.16 (± 0.015)	0.21 (± 0.017)
Prosodic Morpho-syntactic Mimo-gestural	0.22 (± 0.016)	0.19 (± 0.015)	0.25 (± 0.02)

Table 11: F-score, Precision and Recall for feedback prediction for the final model (with all types of feature) and per type of feature: Auto-regressive, Morpho-syntactic, Prosodic, Mimo-gestural, and interaction between Prosodic/Morpho-syntactic/Mimo-gestural features. A margin of error of ± 500 ms around the feedback onset is used.

Our results confirm two hypotheses. Firstly, that taking into consideration all modalities outperforms other feature combinations, in particular unimodal ones. Secondly, the use of detailed linguistic features

also improves the results. Finally, the model reports linguistic information in the significant features that match with our expectations. Moreover, the precision level of the features greatly improves the interpretability of the model and the understanding of feature importance.

As presented in table 11, auto-regressive features play an important role in feedback prediction, by providing information from the context. In our corpus, feedback is produced on average every 5 seconds. Yet, the first two auto-regressive features correspond to a delay shorter than 5s (last feedback between 0-2 and last feedback between 2-5s). One explanation is that feedback could occur close to other feedback depending on the dynamics of the interaction according to the production of important information or specific events. These events can be linked to different salient moments of agreement, elaboration or updating of the common ground during which the listener have to show his/her active contribution. The dynamic also deals with the type of activities (such as storytelling, explanation, etc.) within which the occurrences of feedback can be more concentrated in a short time span (< 5 sec.) (see [Stivers, 2008] or [Bertrand and Espesser, 2017]) than in speech sequences which exhibit more symmetrical speaking times between both participants (less feedback in this case). In this way, as expected from the different studies in the literature, delays longer than 5s are also significant. Consequently, even if feedback is based on a cycle that varies in regularity, the regular baseline shows significantly lower results, indicating that predicting feedback every 5 seconds is not sufficient enough. Information about the delay elapsed since the last feedback produced allows the listener to balance their production of feedback. Moreover, the listener has to adapt this cycle according to the needs of the interaction. As the results by modality of feature show, auto-regressive features alone are not enough on their own to predict feedback, but they do complete information from other modalities.

The three mimo-gestural features stand out as being relevant: smiles, nods and laughs.

The 5 trigrams of tones selected by the model (*SMS*, *BUU*, *DUU*, *LUU*, *DDT*) that are then relevant for feedback prediction mainly correspond to a rising pattern. The latter seems to be the most powerful within the prosodic features. Among the 5 n-grams, 4 correspond to a rising pattern rather associated with a large span in case of the occurrence of an absolute point (B or T). The *SMS* trigram corresponds to a flat intonation in a medium pitch. However, the first tone is of type *Same* (in other words at the same level of the tone preceding the trigram). As a consequence, the intonation variation cannot be determined and could be a falling or a rising depending on the value of the previous tone. The interpretation of this last trigram remains therefore difficult to establish.

Concerning the other prosodic features, we found that silent pauses lasting between 400-600ms and 600-1200ms favor feedback production. This is consistent with different results from the literature: [Cathcart et al., 2003] for a delay of at least 400ms and [Ward and Tsukahara, 2000] for a delay between than 600ms and 1200ms. Finally, punctuation and the number of tokens produced by the speaker are also relevant in predicting feedback.

As for morpho-syntax, 36 n-grams of POS were selected for feedback prediction. 15 n-grams correspond to the end of a chunk, corresponding to a short non-hierarchical syntactic unit [Abney, 1991]. Table B (see Appendix B) shows examples extracted from our corpus. 3 of these n-grams end with an interjection. In French, interjections including phatic markers are often used to involve the listener at the end of a sentence in order to accentuate the previously given information. Moreover, punctuation that is also a significant feature, can mark either the end of a syntagm or the end of a sentence. These results confirm that feedback tends to occur at the end of a sentence. However, we also found significant features that suggested the occurrence of feedback in the middle of a sentence. Our results can be explained by the fact that we considered all the different multimodal feedback and we considered not only generic feedback, as it is the case in most works, but also specific feedback. These results are also consistent with our observation in 4.3, showing that a large quantity of feedback (76%) is produced during the speaker speech. Moreover, 15% of feedback begins during the speech of the speaker and ends in a pause, which indicates that feedback can begin in a middle of a sentence.

15 n-grams contain an adverb or a qualifying adjective (including 7 which also correspond to the end of the chunk). These categories play the role of information modifiers. They complete existing information, which has generally just been given and provides new information. Feedback here can be an explicit mark of grounding.

As for the n-grams that are not explained by the two previous points, 4 contain a common noun (N) and 2 contain interjection. The literature has shown that the category N encourages feedback production. Interjection could again correspond to phatic or discourse markers and maybe also punctuate the previous proposition. Few studies look at feedback produced in overlap, part of our results is explained by the fact that we took into account both speech and non-speech. In this case, the considered POS are those corresponding to the end of syntactic units. Our model learns to predict feedback both in overlap and non-overlap contexts. Relevant n-grams can then occur in the middle or at the end of a syntactic unit, but in specific contexts, for example closing a sequence bringing new information to the listener.

The discussion on the importance of individual features is interesting in order to verify the linguistic consistency of our model. However, one feature alone is not informative enough to trigger feedback, whereas a combination of features may have a strong impact on the probability of its production. The Logit model makes it possible to compute for every such combination of features, the probability of obtaining feedback at a time t . This probability, expressed in $pmean$, can be higher or lower than the probability of obtaining feedback when no features are present in the preceding window. In our dataset, the $pmean$ when no features are present is 0.025, as presented in Table 12. The upper lines of this table show examples of combinations that increase this probability, the lines below illustrating those decreasing the probability. As explained above, the context is highly important but taking into account only auto-regressive features is not sufficient. For instance, if the last item of feedback is produced within the previous 2 seconds (examples 1 and 9), the associated features could totally change the $pmean$. In the configuration of a combination *Last feedback 0-2s, Smile, Punctuation, Determinant Common Noun*, the probability that feedback occurs is almost doubled even if other feedback has just been produced. In contrast, *Last feedback 0-2s, Adverb-Interjection, ML, Punctuation* is associated with a low probability of producing feedback.

Id	Features	Pmean
1	<i>Last.FB.0.2s + Smile + Punctuation + D-Nc</i>	0.046
2	<i>Last feedback 2-5s + Smile</i>	0.038
3	<i>Last feedback 10-20s + Smile + Punctuation + Af + Laugh</i>	0.037
4	<i>Last feedback 10-20s + Pause 600-1200ms + Nod + Af + Punctuation</i>	0.035
5	<i>Last feedback 0-2s + Smile + Nod + Punctuation + I + Va + Vm</i>	0.033
6	<i>Last feedback 0-2s + Vm-R-Pause + Nod + Punctuation</i>	0.031
	<i>No features</i>	0.025
7	<i>Last.FB.0.2s + Vm + Punctuation + Ppn-Vm</i>	0.017
8	<i>Last feedback 2-5s + Nc-S-D + Pause</i>	0.010
9	<i>Last feedback 0-2s + R-I + M.L + Punctuation</i>	0.004

Table 12: Examples of feature combinations observed in the dataset with the probability of obtaining a feedback, expressed in $Pmean$, given by the Logit. When no feature has been produced, the $Pmean$ is 0.025. Examples from 1 to 6 present feature combinations that increase the probability of obtaining feedback. Examples from 7 to 9 present feature combinations that decrease the probability of obtaining feedback. Ngrams of POS are indicated by “-” symbol. Ngrams of tones are indicated by “.”. Abbreviations are detailed in Table A.1 and Table A.2 (see Appendix A)

6. Second Level of Prediction: Predicting Feedback Types

After a first step aiming at predicting positions where feedback may potentially occur, we aim at predicting the type of the feedback in a second step. Recalling that the type concerns the generic/specific distinction proposed in [Bavelas et al., 2000] and the two sub-levels for specific feedback (positive/negative, expected/unexpected), we hypothesized that the different types/subtypes could be triggered by different inviting-cues. We proposed two models for feedback types and sub-types.

6.1 Data Sampling and Model Building

The dataset for feedback type classification is a table containing, for each observed feedback:

- the dependent variable (binary, encoded as 1 for a feedback type and 0 for its opposite type)

- the predictors, in binary type (e.g. presence of a given POS n-gram, presence of a pause, ...) or continuous type (e.g. number of tokens).

Most of the predictors are the same as those presented in the previous section. Unlike the prediction of feedback position, nods, laughs and smiles are extracted in a window of 2 seconds before the onset of the feedback. Positive, negative and concrete tokens were counted since the last feedback. The previous feedback type was implemented (encoded as 0 for generic, 1 for positive-expected, 2 for positive-unexpected, 3 for negative-expected and 4 for negative-unexpected) and was the only auto-regressive feature considered here. Auto-regressive features that used the time elapsed since the last feedback were not kept in this 2nd level of classification, since we considered these features important for the position of the feedback. Overlap was implemented with binary encoding, where 0 indicated that the main speaker was not speaking, and 1 indicated that the feedback was produced in total overlap. We also tested different thresholds of extraction for n-grams of POS and of tones (10, 15, 20, 50, 100). Finally, as in 5.3, we tested the clustering of n-gram of tones and of POS (a total of 164 features for clustered features and 273 for detailed features).

The entire dataset was used for the generic/specific classification. Of course, for positive/negative and expected/unexpected classification, only feedback of specific types were retained.

The question of imbalanced classes is less problematic for this classification than for feedback prediction: the proportion of generic/specific feedback was roughly equivalent (1,206 generic and 1,171 specific). However, the sub-types of specific feedback were slightly imbalanced, especially for polarity: 964 positive and 207 negative feedback. The distribution of (un)expected sub-types was more balanced: 640 unexpected and 531 expected feedback. In the following, we encoded the binary dependent variables as follows: generic/1, specific/0; positive/1, negative/0; unexpected/1, expected/0. In this second level of prediction, the type of the feedback was predicted by using the following classifier equation to convert the *Logit* probability into a binary response:

$$\text{if } (p > p_{\text{threshold}}) \{ \text{response} = 1 \} \text{ else } \{ \text{response} = 0 \} \quad (2)$$

where $p_{\text{threshold}}$ was chosen in such a way that the proportion of feedback of type 1 corresponds to the one observed. This threshold value was estimated based on the training corpus. The model thus provides us with a binary prediction which depends on the predictor values in input.

For each model, a cross-validation was obtained by running a Monte Carlo cross-validation (on 50 trials with a ratio 80%-20% for the training versus the evaluation sample).

A baseline was computed for each level of classification, that randomly predicted class 0 or 1 following the observed distribution.

6.2 Results & Discussion

Results for the 3 models predicting the type of feedback showed that all the best performances were obtained with the features that used detailed n-gram of POS and n-gram of tones. We thus present only these results.

Generic vs. Specific: The best occurrences threshold of n-grams for generic/specific classification is 10. In the case of generic/specific classification, the model returns 30 significant features.

13 features in favor of the *generic* class using auto-regressive, morpho-syntactic and prosodic modalities. 17 features were selected for the *specific* class from all modalities. Performances are presented in table 13 and features importance in Figure C.1 (see Appendix C). The resulting f-score was 0.62. The T-test confirmed that our model is more efficient than the baseline (f-score = 0.48), and always better with all features types ($p \leq 0.01$).

Features modality	F-score	Precision	Recall
All	0.62 (± 0.03)	0.63 (± 0.036)	0.61 (± 0.035)
Auto-regressive	0.58 (± 0.029)	0.59 (± 0.033)	0.58 (± 0.031)
Morpho-syntactic	0.58 (± 0.024)	0.53 (± 0.025)	0.64 (± 0.031)
Prosodic	0.59 (± 0.021)	0.57 (± 0.061)	0.61 (± 0.035)
Mimo-gestural	0.25 (± 0.029)	0.74 (± 0.047)	0.15 (± 0.021)
Baseline	0.48 (± 0.029)	0.48 (± 0.03)	0.49 (± 0.036)

Table 13: F-score, Precision and Recall for feedback generic/specific feedback prediction for the final model (with all types of features) and per type of features: Auto-regressive, Morpho-syntactic, Prosodic, Mimo-gestural. The ground truth used is the type of the observed feedback. The baseline predicts randomly generic/specific feedback based on the distribution observed in the data.

Several interesting observations can be made from the comparison of the features selected in the generic/specific classification. Regarding n-gram of POS, 5 n-grams out of 6 selected for generic prediction correspond to a middle position of a chunk ($\#-S$; $S-\#$; Va ; R ; D). Generic feedback is said to correspond rather to a listening function allowing the main speaker to hold the floor. Feedback can therefore be produced in overlap or in the middle of a chunk. In the same way, the feature overlap favors the prediction of generic feedback.

On the contrary, 5 n-grams out of 8 for the prediction of *specific* feedback corresponds to the end of a chunk or utterance ($D-Nc$; $D-Af$; $Nc-R$; $R-Af$; $Af-I$). The n-gram ($Ppn-Va$) corresponds to a clitic followed by the auxiliary “avoir” (have), and precedes a past participle also corresponding to new information. Finally, 6 of these n-grams contain an adverb or a qualifying adjective ($Nc-R$; $D-Af$; $Af-I$, $Rd-D$, $R-Af$, $R-Ppn$) which play the role of modifying information. As shown in [Bavelas et al., 2017], specific feedback tends to occur after new information. This observation is reinforced by the *pause* features that also promote specific feedback.

At the prosodic level, results show that n-grams of tones selected for predicting *generic* feedback are more often produced in a medium or a small pitch range, or with a flat intonation (SUD , UDU , MS , UDD , DUD). Globally we observed small variations. No points were in a very high or very low absolute peak (B or T). On the contrary, for the prediction of specific feedback, the tones selected showed changes of larger amplitude with tones located in a high pitch and one in a low pitch (LL , TL , UT , SB). Finally, two tones showed a falling intonation (LL , DD).

Nods and smiles did not seem to play a role in predicting the type of feedback. Only laughs are selected by the model. As expected, the presence of laughter promotes the production of specific feedback.

Concerning auto-regressive features, the type of the previous feedback was also selected to predict specific feedback. The interpretation of this observation remains difficult since it is not binary encoded. A possibility is that we added these features in accordance with observations made from our data. Specific feedback can be produced immediately after generic feedback when the semantic content allows it.

In the same way, many items of specific feedback are produced several times depending on the importance of the information given and can be produced with the same type as the previous one.

Positive vs. Negative: The best occurrence threshold of n-grams for positive/negative classification was 20. For the second level of classification, positive/negative classes, 14 features were selected by the model. 7 features from morpho-syntactic, prosodic and mimo-gestural modalities were selected in favor of positive class. 8 features were selected for the negative category, from each auto-regressive, morpho-syntactic and prosodic modality.

Performances are presented in Table 14 features importance in Figure C.2 (see Appendix C). The f-score of 0.36 is very encouraging and outperformed the baseline (f-score = 0.17). For this task too, taking modalities separately lead to a considerable drop in f-score (in particular for auto-regressive

features). Despite the low scores for the prediction of the polarity, features selected by the model tended to support our hypothesis. Indeed, laughs and positive tokens promoted positive feedback and negative tokens seemed significant in predicting negative feedback. We also note that pauses (400-600ms) favor positive feedback where overlap favors negative feedback. One possible explanation is that negative feedback occurs less in the discourse but shows an immediate reaction. Low scores for this task can be partially explained by the highly unbalanced dataset. There is almost 5 times more positive feedback than negative. Future works with wider datasets including different types of interaction (e.g. confrontation, debate) are necessary in order to decide whether this type of classification for specific feedback is relevant for this task.

Features modality	F-score	Precision	Recall
All	0.36 (± 0.049)	0.26 (± 0.04)	0.61 (± 0.092)
Auto-regressive	0.28 (± 0.038)	0.20 (± 0.033)	0.45 (± 0.058)
Morpho-syntactic	0.33 (± 0.048)	0.21 (± 0.032)	0.76 (± 0.011)
Prosodic	0.31 (± 0.029)	0.19 (± 0.022)	0.82 (± 0.054)
Mimo-gestural	0.31 (± 0.035)	0.19 (± 0.025)	0.92 (± 0.043)
Baseline	0.17 (± 0.061)	0.17 (± 0.061)	0.17 (± 0.068)

Table 14: F-score, Precision and Recall for feedback Positive/Negative feedback prediction for the final model (with all types of features) and per type of features: Auto-regressive, Morpho-syntactic, Prosodic, Mimo-gestural. The ground truth used is the type of the observed feedback. The baseline predict randomly generic/specific feedback based on the distribution observed in the data.

Expected vs. Unexpected: The best occurrence threshold of n-grams for unexpected/expected subtypes classification was 20. 30 features were considered as significant by the model without any mimo-gestural features, with 8 for the category *unexpected* from morpho-syntactic and prosodic modalities and 6 features for the category *expected* from auto-regressive, prosodic and morpho-syntactic modalities as presented in Figure C.3 (see Appendix C). Performances in Table 15 showed a higher f-score with only prosodic features. A t-test confirmed that this difference was significant ($p\text{-value} \leq 0.001$). Here, our hypothesis concerning multi-modality is not confirmed: prosodic information alone provided the best f-score. In all cases, prediction always outperformed the baseline (f-score = 0.45).

By considering only the prosodic features selected by the model in respect with the results, we observed that the n-gram of tones selected for the expected category showed a rising and a high pitch span (*HT*).

On the contrary the selected tone for the unexpected category showed a small amplitude with a small falling (*DUD*).

Features modality	F-score	Precision	Recall
All	0.62 (± 0.037)	0.63 (± 0.039)	0.61 (± 0.054)
Auto-regressive	0.55 (± 0.041)	0.66 (± 0.04)	0.47 (± 0.048)
Morpho-syntactic	0.59 (± 0.034)	0.63 (± 0.04)	0.56 (± 0.043)
Prosodic	0.64 (± 0.034)	0.57 (± 0.037)	0.74 (± 0.064)
Baseline	0.45 (± 0.035)	0.45 (± 0.04)	0.46 (± 0.045)

Table 15: F-score, Precision and Recall for feedback Expected/Unexpected feedback prediction for the final model (with all types of features) and per type of feature: Auto-regressive, Morpho-syntactic, Prosodic. The ground truth used is the type of the observed feedback. The baseline predicts randomly generic/specific feedback based on the distribution observed in the data.

7. Conclusion & Discussion

This paper presents an original method for feedback prediction and classification. The first contribution is the identification of the factor complexity in feedback prediction. The second contribution is the elaboration of a hierarchical strategy to investigate both placement and type of feedback. We present two tasks: continuous prediction and feedback type classification. We think this work is original because, to the best of our knowledge, this is the first time that feedback has been considered in its fullest form (generic and specific feedback from vocal/visual and gestural modalities). Despite the complexity of the issue, we aimed to study feedback as universally as possible. For this purpose, we worked on natural conversation. The annotations result from automatic, manual, and manually corrected annotations. Even if nowadays the trend is to have fully automatic annotations, having this wealth of information and level of accuracy allowed us to understand the levels of relevancy of features predicting feedback occurrences and type. Feedback was annotated with multimodal elements and with the generic/specific categories. Sub-classes of specific feedback were experimented in order to test a fine-grained prediction. We used speaker information from prosodic, morpho-syntactic and gestural modalities. We also included information that came directly from the listener behavior, regarding the feedback previously produced (timing and type). We assumed that during natural interaction, interlocutors balance their feedback production in order to provide optimal support to the main speaker, in a sufficient but non-excessive manner. Both feedback functions are produced in this perspective and are regularly distributed. We have tried in this work to group all previous knowledge acquired from the previous studies and to approach the problem in its entire complexity.

Concerning the first main contribution of this article, we would like to recall the main complexity factors at different levels, that we have tried to solve. Based on an extensive literature review from linguistic and machine learning fields, we have attempted to provide a more complete definition of feedback phenomena. On the one hand, we looked at works that define feedback and its inviting-cues. On the other hand, thanks to the various works predicting feedback in continuous and non-continuous ways, we identified key points for feedback prediction and their methodological implication. The first difficulty when predicting feedback is the identification of the phenomenon covered by feedback, which can be variable from one study to another. A clear identification of the task is then necessary. To predict feedback in conversation implies covering all types of feedback, including both verbal/vocal and visual feedback, whether they are produced in overlap or during silent pauses. By its multimodal and functional characteristics, feedback can include a wide set of behaviors. The literature covers different phenomena of feedback production, by typically focusing on one type of feedback (generic, nod or interjection, etc.). Depending on their definition (from the strictest to the broadest definition), the complexity of the feedback prediction task differs significantly. Thus, in the present work, we have taken into account a rich set of multimodal types of feedback (interjection, repetition, longer vocalization, laughs, smiles and nods), classified into two main categories (generic/specific) and into two new sub-categories for specific feedback (positive/negative; expected/unexpected), described in Section 4.1.

The second difficulty relates to the type of data. Indeed, the data used differ in the type of interaction and in the task accomplished by the participants during data collection. Most previous studies used audio-only corpus, and a specific type of interaction such as a map task, doctor/patient dialog or a short narration. In the present work, we used an audio-visual corpus of free face-to-face conversations in order to investigate feedback in the most complete way and to propose a generic model of feedback. Nonetheless, using natural conversation implies *unbalanced data*: feedback instances occur considerably less in interaction than no-feedback instances which causes difficulty in continuous prediction. Moreover, during natural conversation, it is complicated to predict who is currently speaking. This can cause difficulty in training and evaluating models. To overcome this problem, we represented data in a sliding window data-frame for the first level of prediction allowing timing accuracy, every 40ms. We also introduced a new method to reduce the problem of imbalanced classes. We selected only frames where the interlocutor considered as the main speaker had produced more tokens than the interlocutor considered as the listener (expected in the case of feedback production) in the two previous seconds. The aim of this method was to select only moments where the interlocutor considered as the main speaker was then holding-the-floor and consequently this reduced no-feedback instances. This method allowed us to treat both feedback produced in overlap and non-overlap. Concerning the second

level of prediction, a classification with only five labels (generic, positive-expected, positive-unexpected, negative-expected, negative-unexpected) was applied for each item of feedback.

The third difficulty was the features. In the state-of-the-art section (Section 2.2), different modalities have been identified as feedback predictive cues (prosodic, morpho-syntactic, lexical, acoustic, gestural). When using machine learning techniques, it is necessary to identify and select the most informative features to avoid the curse of dimensionality: more features are used, more training data are needed because the number of feature configurations increases. However, the time elapsed between the start of a feature and the time required to trigger a feedback, is not yet very well established. A feature selection work would therefore be of deep importance. We considered it crucial to use information from all modalities to represent the interactions in the most efficient ways. We supported the idea that on the one hand, some features were very important for feedback prediction. On the other hand, we think that the joint occurrence of other cues and their accumulation, also creates an opportunity for feedback. Thus, in the presented research work, we used features from all modalities and we evaluated different quantities of information (set of features) that are given to the model. For this purpose, the performances of different models considering different sets of features, and windows of extraction were compared.

The fourth difficulty resides in the computation of the model. Many techniques for predicting feedback have been proposed in the literature, from handmade (rule-based) to fully automatic methods (using machine learning). The results are difficult to compare given that data, features and evaluation differ according to the study. In our work, we use a Logistic Regression (*Logit*) given the advantages of this method for the considered prediction task. Logit is a machine learning technique which is particularly appropriate for feature selection. This method can handle a large set of features as input and returns a significant subset of features. Another important aspect is the interpretability of the model. The Logit method enables us to compute feature contributions that can be interpreted and compared to a linguistic hypothesis.

Finally, the last and most important difficulty is in dealing with the evaluation of the predictive model where the methods used differ in the literature. Indeed, feedback can involve a lot of behaviors. Interlocutors are able to grasp feedback opportunity, according to the signal produced by the main speaker. However, there is a lot of variability in feedback production. A listener may decide to be more or less expressive in an interaction and has also different options for the expression of his/her feedback. This variability in feedback production (timing and form) renders evaluation difficult. So even if the score is low, it does not mean that the predictive model is not accurate. These problems are raised in a large quantity of works, referred as *expressiveness problem*, for instance in [Morency et al., 2010]. Moreover, most of the studies consider a subset of feedback items (only generic and/or only verbal ones), implicitly reducing the number of feedback items which thus becomes the gold standard. Regarding gold standard, different margins of error have also been used in previous studies. Playing with the margin of error also means easily increased scores (a large margin of error leads to a better score). Our objective here was then to propose potential feedback sites with an acceptable delay. In our research work, we tested 3 different margins of errors inspired from the literature: more or less than 500ms after the feedback onset ([Ward and Tsukahara, 2000]), more or less than 1000ms after the feedback onset, the interval of an observed feedback ([Morency et al., 2010]). Having raised and attempted to address these various difficulties, we also addressed the issue of our second main contribution, i.e. establishing a predictive model based on a new set of multimodal features. We tested and confirmed our 3 hypotheses.

To recap, the first hypothesis concerns the inclusion of high-level and multimodal features to improve the prediction of feedback positions, paving the way for solutions concerning continuous prediction. To this end, we applied a logistic regression to build 4 classification models: *feedback/no-feedback*; *generic/specific*; *positive/negative* and *expected/unexpected*. All models outperformed the different baselines and provided very good results in comparison with other approaches (even though direct comparisons are not possible), reaching an equivalent or higher f-score depending on the evaluation metrics. As stipulated, the importance of multimodal features was confirmed (for 3 out of 4 of models). Moreover, results have shown that all of the models performed better when the more detailed features were used (clustered features vs. detailed features) confirming that high-level linguistic features are more relevant in predicting feedback.

We have therefore identified a new relevant set of multimodal features that allowed us to predict a large set of feedback types, produced both in overlap and in non-overlap speech. Several elements were identified which promote feedback at different levels: rising and large span intonation for the prosodic level, end of chunk, interjection, adverb, qualifying adjective, common noun for the morpho-syntactic level and all mimo-gestural features (nods, laughs and high intensity smiles). Finally, auto-regressive features are important but should be completed by the other features. The second hypothesis, related to the use of a probabilistic model allowing a better understanding and interpretation of the results, in particular by precisely studying feature importance, was confirmed.

Results for the second level of prediction have shown that the predictive features vary depending on the type of the feedback: generic/specific types, positive/negative types and expected/unexpected types, which confirms our third hypothesis. N-gram of POS indicating a middle of a chunk and light intonational variation promotes a generic type. In contrast, n-gram of POS indication the end of a chunk, a modifier (adverb, qualifying adjective), large intonational variation and laughter promote a specific type. Positive tokens, laughter and pauses promote a positive sub-type, where negative tokens and overlap promote a negative sub-type. Pronoun-Verb bi-grams, interjection and small falling promotes unexpected sub-type. Common-noun and rising with a high pitch pan favor expected sub-types. We notice that the finer the prediction becomes, the less information the model needs.

To conclude, our work constitutes the first generic model predicting the possible feedback positions in a conversation as well as their sub-types with this level of precision. Again, let us recall that we predict the *possible* positions, which means that in our reference corpus, not all these positions are filled by feedback (explaining the f-score level).

Our model, by taking into consideration a large set of features, also constitutes an important descriptive contribution of the nature of feedback.

Henceforth, several perspectives for improvement can be drawn. First, we are now working on the integration of new features and feedback elements into the model: gaze direction [Bavelas et al., 2002, Ozkan and Morency, 2010, Hjalmarsson and Oertel, 2012]; blinking [Hômke et al., 2017], eyebrow movements and facial expressions. We will also work on different feature aggregation solutions. A second type of work will concern the use of embedding and associated learning techniques, moving towards neural networks.

Moreover, it is necessary to complete objective evaluations with perceptual experiments. In particular, we have observed an important improvement by using a margin of error of $\pm 1000\text{ms}$ (f-score = 0.49). We will verify the validity of this window by looking for the accuracy of the feedback thanks to a perceptual experiment consisting in presenting the participant feedback produced within or outside the evaluated windows.

Finally, we will also integrate the question of the modality of the feedback (vocal, visual and bimodal) by studying whether it plays a role in the model (in other words whether some features can be associated with different feedback modalities) and more generally whether it is possible to predict the feedback modality together with its type. [Truong et al., 2011] showed that different inviting cues reside between vocal and visual modalities. In this work, we considered the site and the type of feedback as a priority but we could also include in our future work the question of modality.

8. Acknowledgments

This work, carried out within the Institut Convergence ILCB (ANR-16-CONV-0002), has benefited from the support of the French government, managed by the French National Agency for Research (ANR) and the Excellence Initiative of Aix-Marseille University (A*MIDEX).

Appendix A. POS and Tones abbreviation

POS	Type	Example	POS	Type	Example
Af	Qualitative adjective	petit	Rq	Other adverb of negation	que
A-	Other adjective	aucune	R-	Other adverb	facilement
Cc	Coordinative Conjunction	et	Sa	Preposition "à"	à
Cs	Subordinating Conjunction	lorsque	Sd	Preposition "de"	de
I	Interjection	hein	SP	Preposition + Determiner	du
D-	Determiner	le	S-	Other preposition	dans
Nc	Common noun	chapeau	Van-	Auxiliary avoir infinitive	avoir
Np	Proper noun	Baudelaire	Vapp	Auxiliary avoir present participle	ayant
Nk	Cardinal noun	huit	Vaps	Auxiliary avoir past participle	eu
Ppn	Personal pronoun nominative	je	Va-	Other form of auxiliary avoir	ai
Ppj	Personal pronoun accusative	la	Ven-	Auxiliary être infinitive	être
Ppd	Personal pronoun dative	lui	Vepp	Auxiliary être present participle	étant
Ppo	Personal pronoun oblique	moi	Veps	Auxiliary être past participle	été
Pr	Relative pronoun	qui	Ve-	Other form of auxiliary être	suis
Pd	Demonstrative pronoun	ce	Vmn-	Main verb infinitive	aimer
Pi	Indefinite pronoun	personne	Vmpp	Main verb present participle	finissant
Pt	Interrogative pronoun	quel	Vmps	Main verb past participle	payé
Px	Reflexive pronoun	se	Vm-	Other form of Main verb	partirent
Ps	Possessive pronoun	le mien	U	Unknown or foreign word	zarbi
Pk	Cardinal pronoun	quinze	Wd	Strong punctuation	.
Rn	Particle of negation	ne	Wm	Weak punctuation	,
Rd	Adverb of negation	pas	#	Pause	

Table A.1: The encoding of the French unitary POS (Part-Of-Speech) entering the definition of morphosyntactic bigrams and trigrams.

Absolute Value	Encoding	Relative Value	Encoding
Top	T	Higher	H
Bottom	B	Lower	L
Mid	M	Same	S
		Upstepped	U
		Downstepped	D

Table A.2: The encoding of the MOMEL anchor points of the MOMEL-INTSINT [Hirst, 2022].

Appendix B. First Level of Prediction: Predicting Feedback Positions

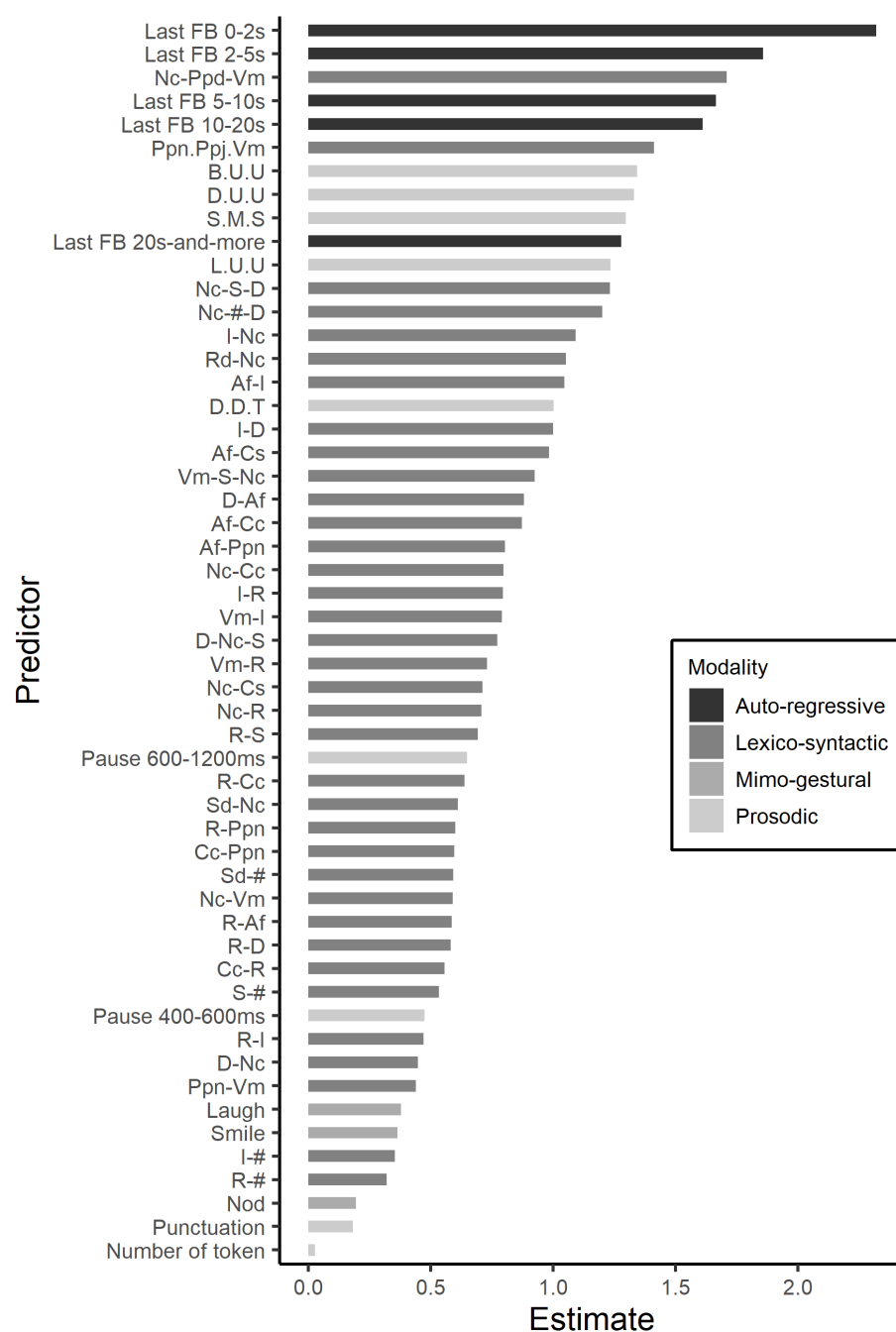


Figure B: Feature importance for **feedback** prediction. Ngrams of POS are indicated by “-” symbol. Ngrams of tones are indicated by “.” symbol. Abbreviations are detailed in Table A.1 and Table A.2 (see Appendix A).

Ngrams	Examples
Vm-R	<i>Je me suis dit c'est trop</i>
Rd-Nc	<i>Ca pourrait être pas mal</i>
Vm-I	<i>On verra hein</i>
Af-I	<i>J'étais présente hein</i>
Sd-Nc	<i>J'ai peur qu'on ait beaucoup de boulot</i>
R-Af	<i>Les calanques de Marseille c'est vraiment beau</i>
D-Nc	<i>Pour apprendre la langue</i>
Nc-Vm	<i>Je voulais attendre que ma soeur revienne</i>
R-I	<i>C'est pour ça que personne n'y va en fait hein</i>
Ppn-Ppj-Vm	<i>C'est pareil parce que je la voyais</i>
Vm-S-Nc	<i>Ouais je voulais pas aller en master</i>

Table B: Examples of bigrams and trigrams that can mark the end of a chunk. Ngrams of POS are indicated by “-“ symbol. Abbreviations are detailed in Table A.1 and Table A.2 (see Appendix A).

Appendix C. Second Level of Prediction: Predicting Feedback Types

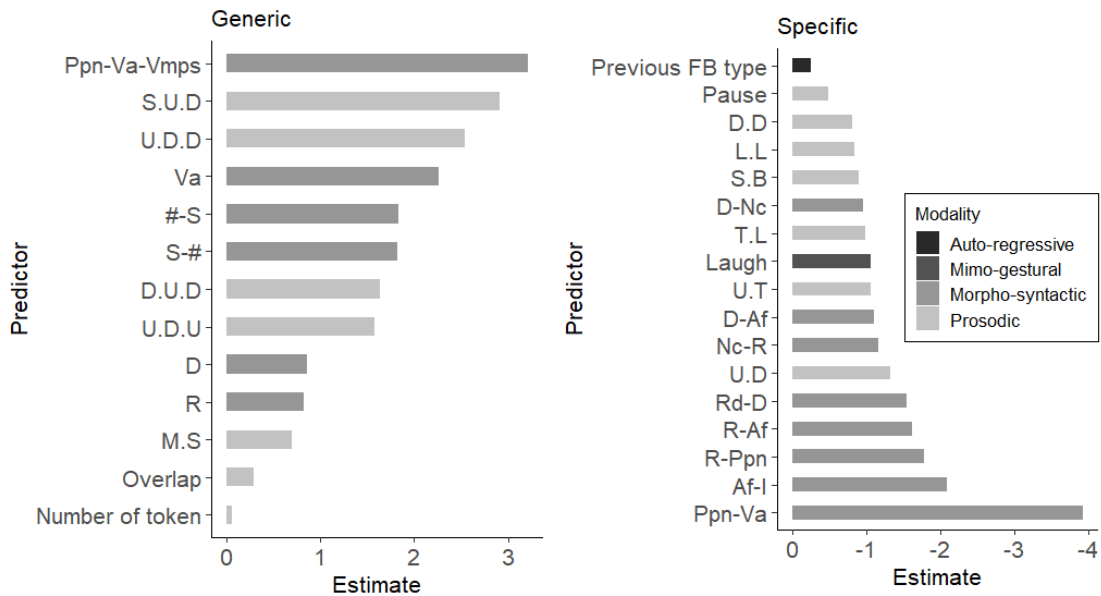


Figure C.1: Feature importance for the **generic** (left panel) and **specific** (right panel) feedback classes. Ngrams of POS are indicated by “-“ symbol. Ngrams of tones are indicated by “.” symbol. Abbreviations are detailed in Table A.1 and Table A.2 (see Appendix A).

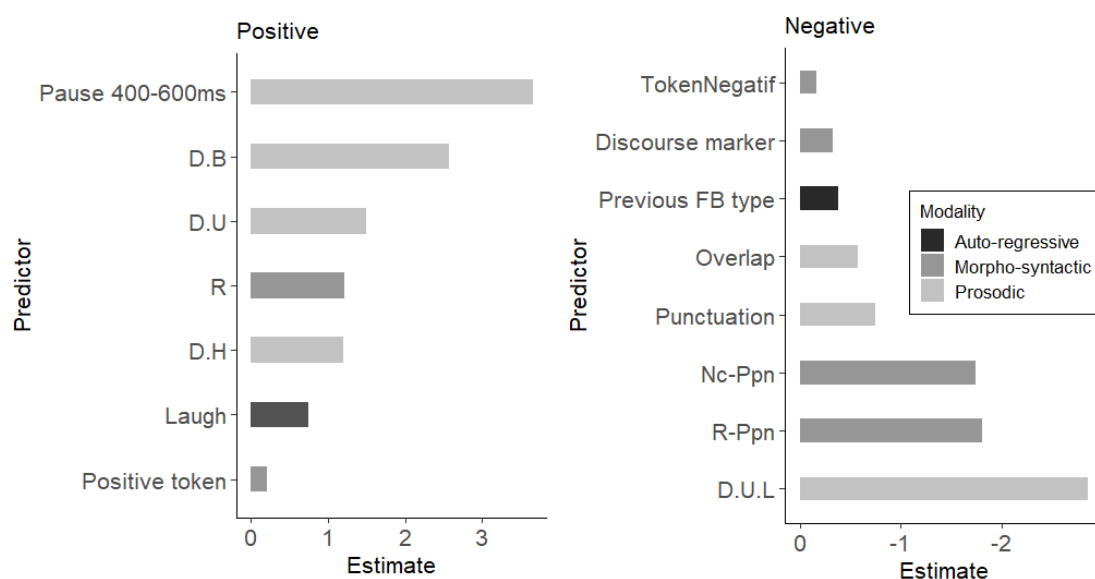


Figure C.2: Feature importance for the **positive** (left panel) and **negative** (right panel) specific feedback sub-classes. Ngrams of POS are indicated by “-” symbol. Ngrams of tones are indicated by “.” symbol. Abbreviations are detailed in Table A.1 and Table A.2 (see Appendix A).

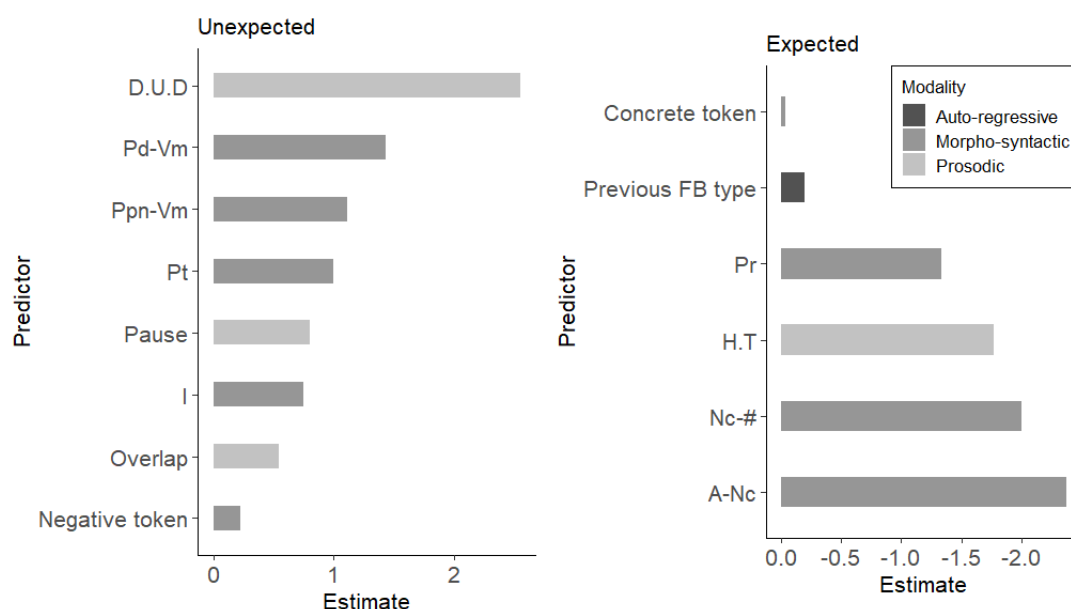


Figure C.3: Feature importance for the **unexpected** (left panel) and **expected** (right panel) specific feedback sub-classes. Ngrams of POS are indicated by “-” symbol. Ngrams of tones are indicated by “.” symbol. Abbreviations are detailed in Table A.1 and Table A.2 (see Appendix A).

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